

FloodLAMP

Biotechnologies, PBC

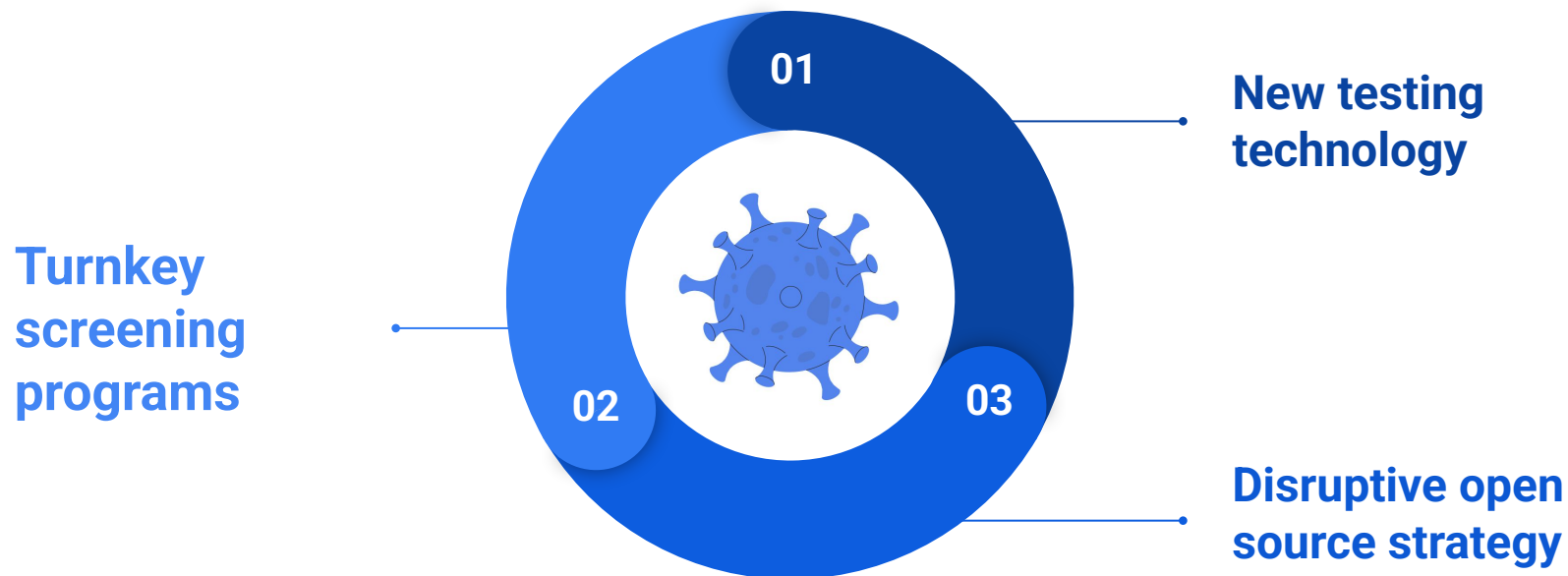
Extending the Reach of LAMP:

A platform for globally accessible disease screening

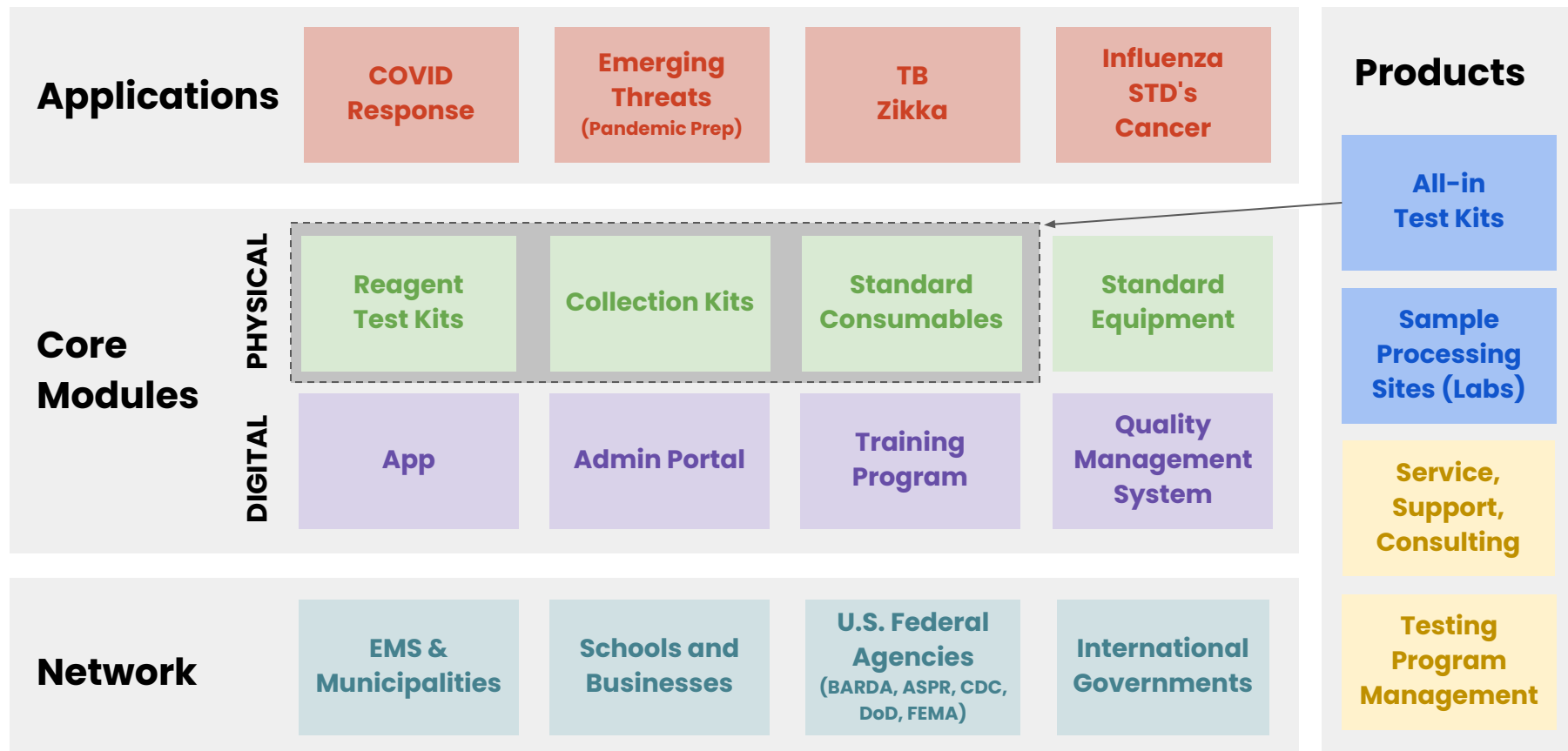
BARDA Market Research – March 7, 2022

Randy True, Founder and CEO | randy@floodlamp.bio

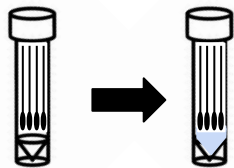
“FloodLAMP's mission is to improve global health and resiliency through universal access to rapid molecular testing.”



2) Platform – Wraparound program components



PCR and LAMP Sister Assays



Streamlined Sample Prep

- Standard Rabe-Cepko TCEP/EDTA inactivation solution
- Added directly to dry swabs
- Same sample for both tests

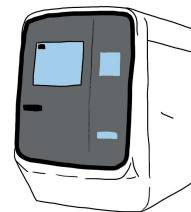
- Same sample for both tests
- 2 μ L sample volume



visual readout

QuickColor™ LAMP Test

- M1804 MM
- Standard by-the-book 25 μ L reaction
- 3 primer sets mixed (AS1e, N2, E1)
- 25 minutes at 65°C

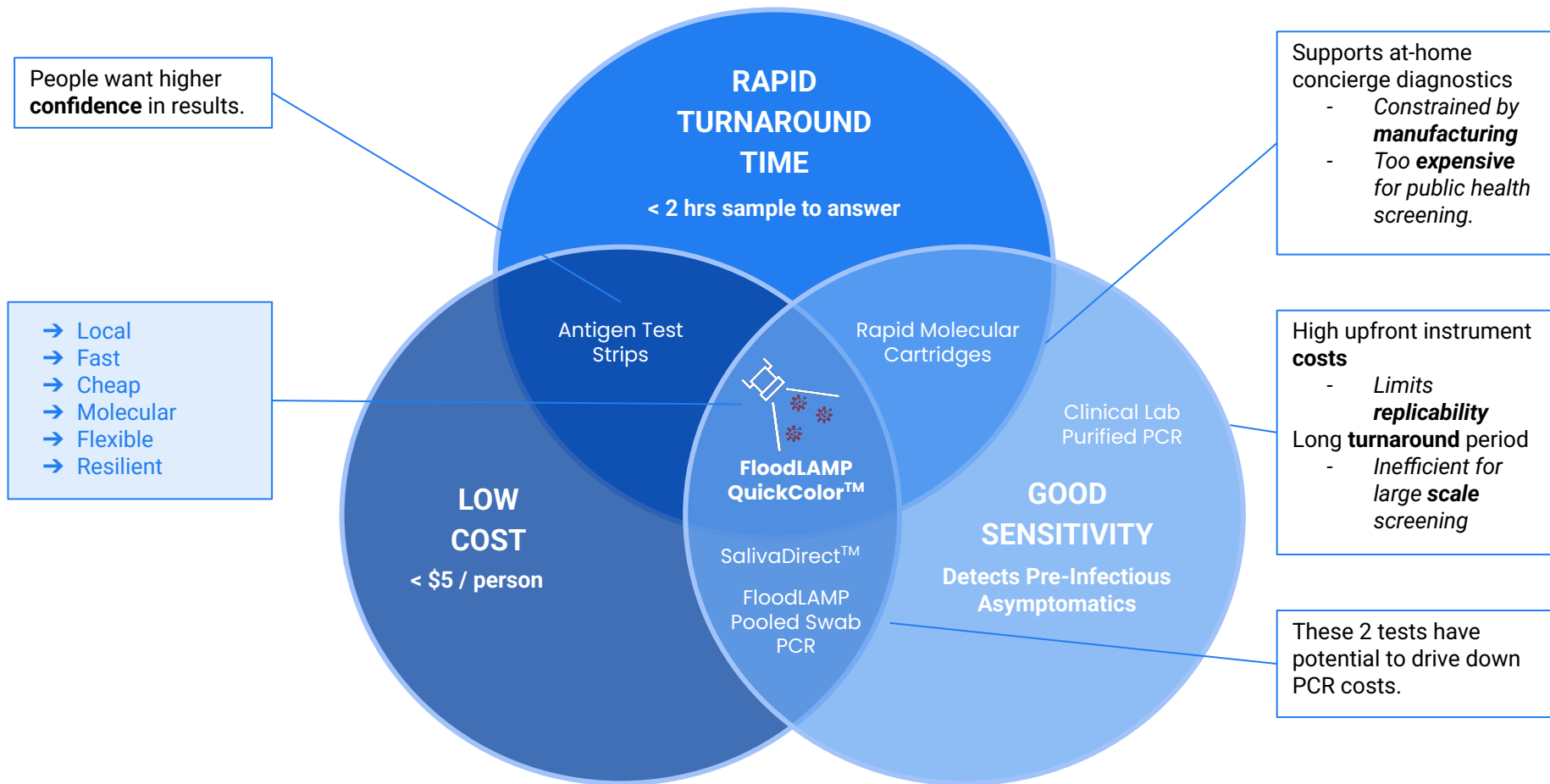


Standard RT-qPCR instrument

EasyPCR™ Test

- Luna E3006 kit
- Standard by-the-book 20 μ L reaction
- CDC primers in SalivaDirect config (N1, RNaseP)

The Performance Sweet Spot



Addressing Current Vulnerabilities

1

Antigen Tests are at Risk of Failure with New Variants

2

Multi-Gene Molecular Tests are Resilient to New Variants

Test combines primer sets for 3 genes

→ ORF1ab ; N2 ; E1

LAMP uses 6 primers and is highly tolerant to mismatches

→ (see recent *NEB preprint analyzing our primers*)

3

Reagent Based Molecular Tests are Adaptable and Scalable

Primers swapped quickly and cheaply.

→ Other test components remain the same.

Primers produced in large volumes and distributed quickly.

→ 10s of M of reactions / production run.

4

FloodLAMP can deliver Prepositioned Assets

For rapid response, scaling & surge capability

Our systems comprise of ordinary, low-cost lab equipment

→ Offering best-in-class capacity per cost

We deliver custom-designed wraparound components

→ E.g: digital tools, QA, training

Operating Programs in the Real World



EMS Leadership Conference Screening

→ 1,000 attendees



Municipal & EMS Deployments

- Coral Springs, FL approx. 132,000
- Davie, FL approx. 104,000
- Bend, OR approx. 94,000

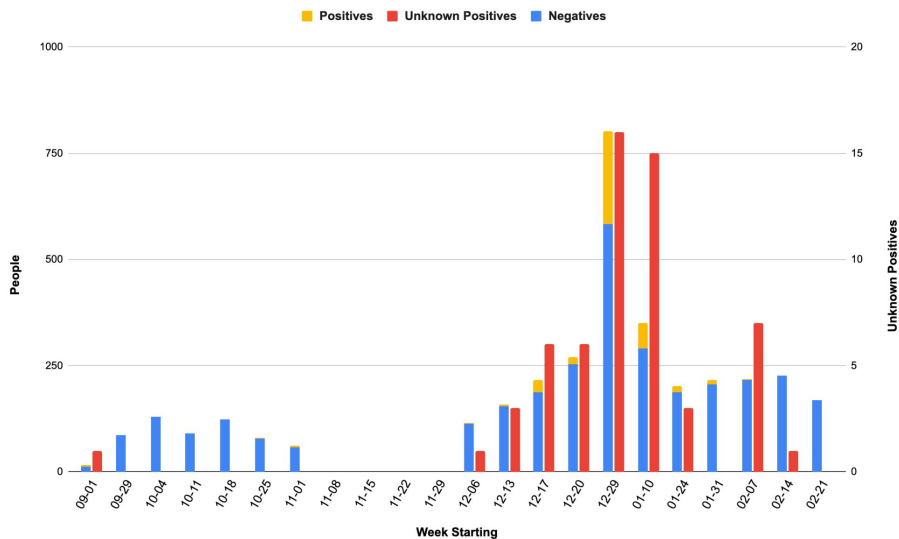


Preschool Screening with Family Pooling

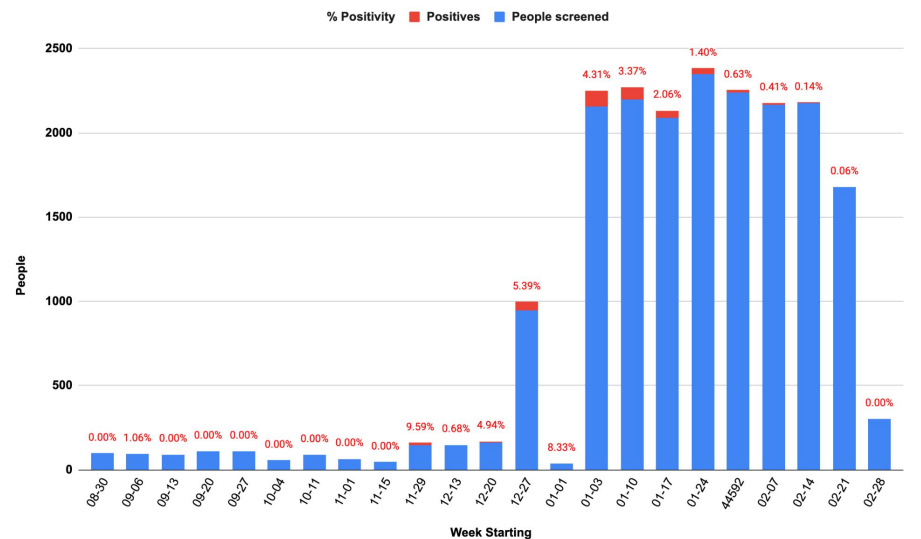
- Approx. 40 families
- 65-105 people per bi-weekly screening

FloodLAMP Surveillance Testing Stats			Thru Mar 2, 2022
Org	Pools	People	Positives
FL FF	1,107	1,779	44
FL FTFC	61	195	0
Kent Summer Camp	190	696	0
Coral Springs City	7,400	21,676	348
Davie Fire Rescue	2,235	4,226	55
Pink Shoes Production	671	671	1
Bend Fire Rescue	617	617	215
TOTAL	12,281	29,860	663

FloodLAMP's successful response to surges

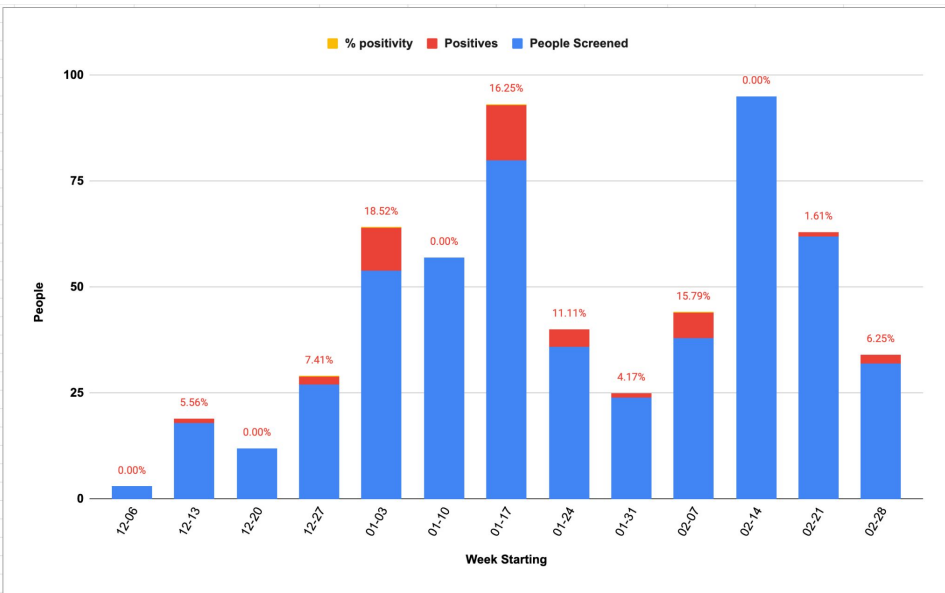


Town of Davie, FL

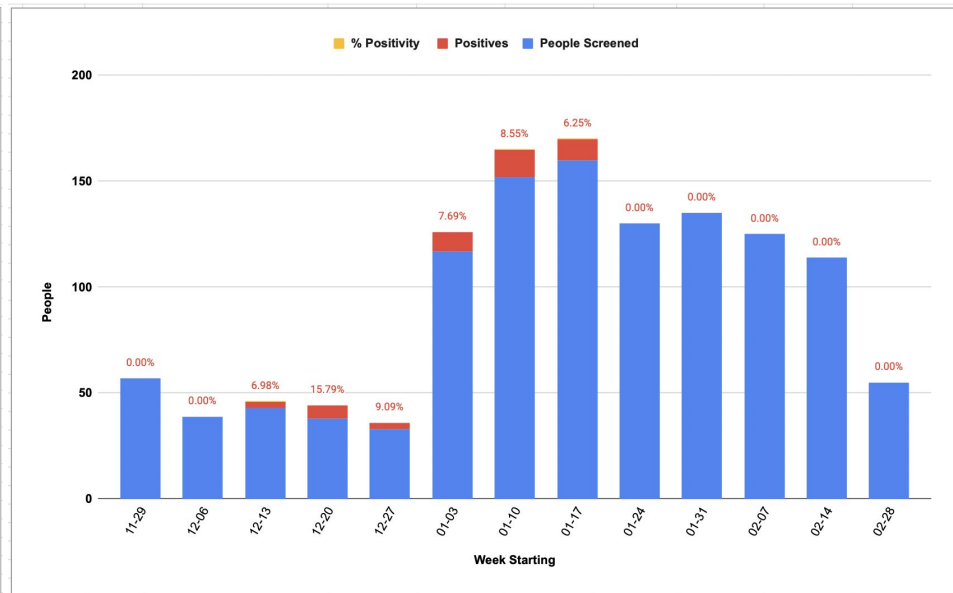


Coral Springs, FL

FloodLAMP's successful response to surges



Bend, OR



Preschool, CA

Case Report: Pre-School Screening

- Several infections detected **1-2 days** before symptoms or antigen tests
- Family pooling, in 3 cases, detected infections among student family members and **protected spread in the congregated setting** (such as schools and workplaces).

	1/2/22	1/3/22	1/4/22	1/6/22	1/7/22	1/8/22	1/9/22	1/10/22	1/13/22	1/14/22	1/17/22
	0	1	2	4	5	6	7	8	11	12	15
Family Pool	POS FloodLAMP										POS FloodLAMP
Parent 1	Asymptomatic										
	POS FloodLAMP (Family Pool)	POS FloodLAMP (Indiv)					POS FloodLAMP (Indiv)	> POS FL Result			POS FloodLAMP (Family Pool)
	NEG Antigen									NEG Antigen	
			POS PCR Lab >	> > > >	> POS PCR Result						
Parent 2	Asymptomatic		Symptoms								
	POS FloodLAMP (Family Pool)	POS FloodLAMP					POS FloodLAMP (Indiv)	> POS FL Result			POS FloodLAMP (Family Pool)
	NEG Antigen									NEG Antigen	
		POS PCR Lab >	> POS PCR Result								
Child 1	Asymptomatic		Symptoms	Asymptomatic							
	POS FloodLAMP (Family Pool)	NEG FloodLAMP					POS FloodLAMP (Indiv)	> POS FL Result			POS FloodLAMP (Family Pool)
			POS Antigen							NEG Antigen	
					POS PCR Lab >	> POS PCR Result					
Child 2	Asymptomatic		Symptoms	Asymptomatic							
	POS FloodLAMP (Family Pool)	NEG FloodLAMP					POS FloodLAMP (Indiv)	> POS FL Result			POS FloodLAMP (Family Pool)
			POS Antigen						POS Antigen (slight	NEG Antigen	
				POS PCR Lab >	> POS PCR Result						

Case Report: New Year's Eve

Detection of an **asymptomatic** infection that was **missed** by **3 expensive** OTC/POC rapid molecular **tests**

- *Detect, Lucira, Mesa Accula* were negative
- *BinaxNOW* wasn't positive until 2 days later.
- All *FloodLAMP* positive samples confirmed by *PCR*

FloodLAMP New Years Eve Case Report						
caught asymptomatic infection that all antigen and OTC/POC molecular tests missed						
FloodLAMP EasyPCR on saliva				POS - SalivaDirect PCR Ct 27.7 8 PM	POS - SalivaDirect PCR Ct 29.0 11 AM	
Accula Rapid Molecular PCR (Worksite)					Accula NEG - nasal swab 4 PM	
Lucira Rapid Molecular LAMP				Lucira NEG - nasal swab 5 PM		
Detect Rapid Molecular LAMP	NEG - nasal swab 10 AM		Detect NEG - nasal swab 2 PM			
CaseStart Antigen						
FlowFlex Antigen		NEG - nasal swab 8 AM	NEG - nasal swab 2 PM			POS - nasal swab 10 AM
BinaXNow Antigen				NEG - nasal swab 5 PM	NEG - nasal and throat swabs 10 AM	POS - nasal swab 10 AM
FloodLAMP QuickColor LAMP Nasal Swab		NEG - nasal swab 1 PM		NEG - nasal swab PCR Ct Und/39.4 5PM	POS - nasal swab PCR Ct 32.2 10 AM	
FloodLAMP QuickColor LAMP Throat Swab			POS - nasal swab PCR Ct 33.9 9 AM	POS - throat swab PCR Ct 30.2 5 PM	POS - throat swab PCR Ct 35.5 10 AM	
	12-27-21	12-29-21	12-31-21	12-31-22	1-1-22	1-2-22

Open EUAs

- Provide a path to **establishing generics in diagnostics** (*direct opposition to typical IVD EUA*).
- FloodLAMP has submitted **2 Full EUAs + pre-EUA** as open protocol.

	 Typical IVD EUA	 CDC EUA	 SalivaDirect™	 SHIELD	 FloodLAMP EasyPCR™	 FloodLAMP QuickColor™
Disclosure of all chemicals and reagents?	No	Yes	Yes	Yes	Yes	Yes
Chemical and reagents available from multiple vendors?	No	Yes	Yes	No	Yes	Yes
Disclosure of primer sequences?	No	Yes std for PCR	Yes	No ProptryThermo	Yes	Yes
Primers commercially available from multiple vendors?	No	Yes	Yes CDC Primers	No ProptryThermo	Yes CDC SD Primers	Yes Available but not launched
Supply chain robust?	No	No/Maybe	Yes	No/Maybe	Yes	Yes
EUA Sponsor Organization Type	For Profit Company	Govt	Academic Not for Profit	Academic Not/For Profit ?	Public Benefit Corp	Public Benefit Corp
Designation of CLIA labs	Kit Sales	N/A open RoR	Impact & Expansion	Impact & Expansion	Impact & Expansion	Impact & Expansion

Support for Further Implementation

Federal support is required for rapid deployment to both a number of interested organizations (e.g. EMS, early child-care hospitals) and to the communities most in need.

Rapid Response Surge Screening

- Immediately deployable to hotspots
- Highly replicable with low capital cost (*unlike most PCR based mobile testing*)
- Partner with local community groups for building trust and driving adoption

Sustained Community Assurance Screening

- Anchored in schools and workplaces
- Expand to entire communities with flexible and convenient options
- Marry with policies such as test-to-stay that drive adoption

Multi-Site Study

Compare to antigen studies on key endpoints:

- Additional reduction in Rt from Screening Program
- % participation in program
- Participant satisfaction in program
- School/work days saved by program (test-to-stay)

Network of EMS Surveillance Screening Sites

High need for screening critical first-responders owing to the large degree of vaccine hesitancy in this population

- FloodLAMP has established relationships with EMS leadership (*facilitates rapid expansion*)
- Ideal group to expand screening to schools and businesses (*EMS has community trust*)

FloodLAMP Team



Randy True
Founder & CEO

- Founder TMI Inc. - Acq. Affymetrix
- Former VP of R&D at Affymetrix

bit.ly/biosketch-randalltrue

co-author global LAMP consortium review: bit.ly/gLAMP-review
<https://abrf.memberclicks.net/jbt-2021-september-issue>



Kevin Schallert
COO

- Co-Director covid19sci.org, National Volunteer Scientist DB
- Former Founder & COO of VineEye



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UX/Design Lead



Gary Withey, Ph.D.
Vice President of Research & Development



Dr. Peter Antevy, MD
Program Medical Director



Brandon Smith
Lab Assistant



Esmé Thornhill-Davis
Communications Intern

Scientific Advisory Board

Anne Wyllie

Yale, Lead Researcher – SalivaDirect

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UC San Francisco, Genoa Ventures

Industry Advisors

Tim Lugo

William Blair Biotechnology Group Head

Zarak Khurshid

Asymmetry Capital, Top MDx Analyst

John Edge

Oxford Internet Institute, Blue Field Labs, ID2020

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Research-Aid Networks: Jeremy Rossman
EMS deployments: Adam, Danny, Alex, Petar, Dr.
Paul Pepe
*Thanks too to all of our advisors, volunteers &
former employees!!*

APPENDIX

Quad Chart

Objective:

FloodLAMP's mission is to improve global health and resiliency through universal access to rapid molecular testing. We have developed a key capability that can help realize the enormous potential of the field of diagnostics in ending this pandemic, preparing for the next one, and improving human health globally. Our near term objective is to deploy hundreds of sites delivering easy to use, flexible, fast turnaround COVID surveillance screening to drive durable adoption in the communities most in need.

Description of effort:

- Fully integrated turnkey surveillance programs comprising all equipment, consumables, test kits, pooled collection kits, digital tools, training, and support;
- Systems cost of \$2K gives 20K/day capacity, at low cost of <\$2 pp (incl labor);
- Currently 8 systems deployed with EMS departments in 3 states, 11 staff trained to run test (5 with no experience), 5,600 people screened, 36 unknown positive cases detected, 44 known confirmed (100%), no known FN;
- Full FDA EUAs submitted for 2 tests, duplex PCR and colorimetric LAMP, as open source protocol EUAs, clinical evaluation by Stanford CLIA lab with 98% (PCR) and 90% (LAMP) sensitivity, and 100% specificity, pre-EUA sub for pooled home collection w/ FloodLAMP Mobile App, IRB approved.
- Key advisors, collaborators and industry connections (Anne Wyllie of SalivaDirect, Robby Sikka's Sports & Society call, gLAMP consortium, NEB, LGC)

Benefits:

- Scalable and replicable high quality molecular screening, deployable anywhere;
- Rapid response to hot spots: 10K/day capacity bringup in 24hrs;
- Outbreak risk reduction and disease suppression in high-need communities;
- Acceleration of return-to-normal and economic activity;
- Increased adoption over DIY antigen tests through convenience to end user;
- Improved mitigation effectiveness of pooling at the family/household level;
- Establishment of critical new regulatory paradigm – the generics of diagnostics through open source protocol FDA authorized tests, LAMP primer standards, and point-of-need flexible public health testing.

Challenges:

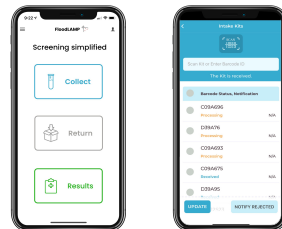
- Regulatory barriers – access to FDA review as a startup;
- Supply chain and staffing;
- Educating the public and political leaders of the benefits of frequent screening.

Maturity of Technology:

- Demonstrated real world success of full programs at multiple sites;
- Test system clinical evaluation successful and FDA EUA validations completed;
- Patent filings and licenses in place to secure control of key IP;
- Pilot manufacturing of test reagents (40K reactions/day);
- Reagent supply partners in place (LGC Biosearch and New England Biolabs);
- Quality system and training program in development.

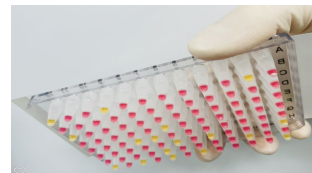
FloodLAMP Mobile App:

- At home collection of family pools;
- Lightweight LIMS for tracking and reporting;
- Admin for mgmt, results, and metrics.



QuickColor™ COVID-19 Test and Turnkey Screening Program:

- Instrument free rapid molecular test,
- 2 hrs to setup and validate site,
- 10K+/day throughput at <\$5K cost.



Major Milestones:

Milestone 1 - State level contract for all EMS sites screening first responders:

- Hub and spoke model for smaller departments;
- Pilot expansion to under served communities, schools and workplaces.

Milestone 2 - FDA open source protocol EUAs for pooled direct LAMP and PCR:

- Complete asymptomatic clinical trial with enrichment strategy;
- EUA for DTC pooled collection kit using FloodLAMP Mobile App;
- Quality and training systems completed;
- U.S. FDA EUA enables global distribution and copy cats.

Milestone 3 - Full suite of open source resources:

- Modern website with all information, tools, products for various scales;
- Includes quality and training systems.

Milestone 4 - 5M tests deployed:

- Key reagents are in-hand, ready to kit and distribute.

Proposed Funding:

- Phase 0: 1 week – deep dive due diligence;
- Phase 1: 2-4 weeks – workplan incl Milestone 1 and TBD based on feedback;
- Phase 2: 2-6 weeks – workplan incl Milestones 1, 2, 3 and TBD based on feedback.

Filling a Gap in the Testing Landscape



Current testing paradigms cannot stem a pandemic caused by an asymptotically transmitted, aerosolized pathogen.



- Scaling a single test up to very high levels offers **opportunities for innovation** in sampling, test chemistry, and program configuration.
- A "**Point of Need**" modality of testing is needed that combines the scalability and low cost of liquid reagent processing with the ease, flexibility and decentralized nature of POC/At-Home.
- **Public Health processing sites** can fill the gap between CLIA labs and DIY at home tests.
- Want this new capability to be **additive** and not cannibalize the clinical diagnostic infrastructure.

Open EUA's – 2 Full Submissions + Pre EUA

FloodLAMP QuickColor™ COVID-19 Test

Instructions for Use v1.2

IVD

COVID-19 Emergency Use Authorization Only
For *in vitro* diagnostic (IVD) Use

www.floodlamp.bio
FloodLAMP Biotechnologies, PBC | 930 Brittan Ave. San Carlos, CA 94070 USA

FloodLAMP EasyPCR™ COVID-19 Test

Instructions for Use v1.1

IVD

COVID-19 Emergency Use Authorization Only
For *in vitro* diagnostic (IVD) Use

www.floodlamp.bio
FloodLAMP Biotechnologies, PBC | 930 Brittan Ave. San Carlos, CA 94070 USA



**FloodLAMP
Biotechnologies**

A Public Benefit Corporation

**FloodLAMP Pooled Swab
Collection Kit DTC**

For use with the FloodLAMP
Mobile App

Open EUA – Disclosure of Components

FloodLAMP QuickColor™ COVID-19 Test
IVD FDA Emergency Use Authorization - Instructions for Use (v1.2)

Table 1: Validated reagents used with the Test

Item	Concentration	Chemical Composition	Vendor	Catalog Number
TCEP	.5 M	tris(2-carboxyethyl)phosphine hydrochloride	Sigma-Aldrich Millipore Sigma	646547-10X1ML
EDTA	.5 M	Ethylenediaminetetraacetic acid	Thermo Fisher	15575020
NaOH	10 N	Sodium Hydroxide	Sigma-Aldrich	SX0607N-6
Nuclease-free Water		Ultrapure Water, nuclease-free	Thermo Fisher	10977015
NaCl	5 M	Sodium Chloride	Thermo Fisher	24740011
Guanidine HCl	6 M	Guanidine Hydrochloride	Sigma-Aldrich	SRE0066
Colorimetric LAMP MM*		Colorimetric LAMP Master Mix	New England Biolabs	M1804

* Item may not be substituted for equivalent. Only the specified vendor and catalog number may be utilized.

The FloodLAMP QuickColor™ COVID-19 Test uses 18 LAMP primers targeted for 3 different SARS-CoV-2 genes, with 6 primers for each target. All 18 primers are mixed together and are input into a single amplification reaction. Primer names and sequences are listed in Table 2. Primers may be purchased pre-blended from the vendor LGC Biosearch Technologies with the product names LAMP_S2-As1e, LAMP_S2-N2, LAMP_S2-E1. Alternatively, primers may be purchased as individual custom oligos. Appropriate validation of primer mixes from custom oligos is required. See Primer Preparation below for more information.

Table 2: Primer names and sequences

Primer Name	Sequence (5'-3')
ORF1ab gene (AS1e)	
Orflab_FIP	TCAGCACACAAAGCCAAAAATTTATTTCTGTGCAAGGAAATTAAGGAG
Orflab_BIP	TATTGGTGGAGCTAAACTTAAAGCCTTTTCTGTACAATCCCTTTGAGTG
Orflab_F3	CGGTGGACAAATTGTAC
Orflab_B3	CTTCTCTGGATTAAACACTT
Orflab_LF	TTACAAGCTTAAAGAATGTCTGAACACT
Orflab_LB	TTGAATTTAGGTGAACATTGTGCACG
N Gene (N2)	
N2_FIP	TTCCGAAGAACGCTGAAGCGGAAGTATTACAAACATTGGCC
N2_BIP	CGCATTGGCATGGAAGTCACAAATTTGATGGCACCTGTGTA
N2_F3	ACCAGGAACATAATCAGACAAG
N2_B3	GACTTGATCTTTGAAATTTGGATCT
N2_LF	GGGGGCAAAATGTGCAATTTG
N2_LB	CTTCGGGAACGTGGTTGACC

Table 7: Primer-Guanidine Solution

Component	Volume (1 reaction)	Volume (1 reaction x 100) 1 x 96-plate w/ 4% overage
10X LAMP Primer Mix	2.5 µL	250 µL
Guanidine HCl (400 mM)	2.5 µL	
Guanidine HCl (6 M)		16.7 µL
Nuclease-free Water	5.5 µL	783 µL
TOTAL VOLUME	10.5 µL	1050 µL

Table 8: Colorimetric LAMP Amplification Reaction

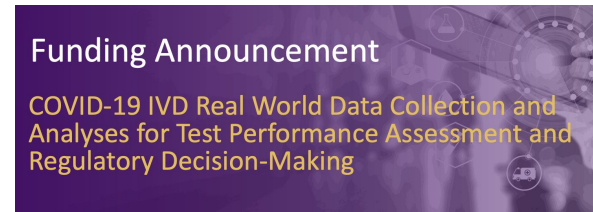
Component	Volume (1 reaction)	Volume (100 reactions)
Primer-Guanidine Solution	10.5 µL	1050 µL
Colorimetric LAMP MM	12.5 µL	1250 µL
SUBTOTAL VOLUME	23 µL	2300 µL
Sample	2 µL	
REACTION VOLUME	25 µL	

Push for Regulatory Progress

Prepare for and Respond to Existing Viruses, Emerging New Threats, and Pandemics Act (PREVENT Pandemics Act)

Sec.505.Facilitating the use of real world evidence.	Requires FDA to issue or revise guidance on the use of real-world data and real-world evidence to support regulatory decisionmaking, including with respect to real-world data and real-world evidence from products authorized for emergency use.
Sec.506.Advanced platform technologies.	- Creates an advanced platform technology designation to expedite the development and review of new treatments and countermeasures that use cutting-edge, adaptable platform technologies that can be incorporated or used in more than one drug or biological product. - Requires FDA to issue guidance on the implementation of the new designation.
Sec.507.Increasing EUA decision transparency.	Provides FDA with authority to share more safety and effectiveness information with the public about products authorized for emergency use.
Sec.508.Improving FDA guidance and communication.	- Requires publication of a report identifying best practices across the FDA and other applicable agencies for the development, issuance, and use of guidance documents and for communications with product sponsors and other stakeholders, and a plan for implementing such best practices. - Requires FDA to publish a report on the agency's best practices for communicating with medical product sponsors and other stakeholders, and a plan for implementing such best practices.
Sec.509.GAO study and report on hiring challenges at FDA.	Directs GAO to issue a report assessing FDA's hiring, recruiting, and retention practices, policies, and processes, and their impact on FDA's ability to carry out its public health mission, particularly in light of the COVID-19 pandemic.

Subtitle B—Mitigating Shortages



<https://reaganudall.org/research-funding-opportunity>

LAMP EUAs for SARS-CoV2

EUA Date	Test	Diagnostic	Target	IC	Detection	Extraction	Amplification	LOD spike	LOD GE/ 3ml swab
Nov-17	Lucira (At Home POC)	Lucira COVID-19 All-In-One Test Kit	N, N	External + IC	Colorimetric	~Direct	Lucira device	Inactivated virus	2,700
Oct-05	Season (2 duplex)	AQ-TOP COVID-19 Rapid Detection Kit PLUS	orf1ab, N	Human RNaseP	PNA Probe	Manual (Qiagen_60704 or Season_5S-1300) or Automated (Panagene_PNAK-1001 on PanaMax48)	BioRad_CFX96 or ABI7500	NCCP_43326 genomic RNA	3000
Sep-01	Detectachem	MobileDetect Bio BCC19 (MD-Bio BCC19) Test Kit	N, E	only external controls	Colorimetric	Direct (1µl of transport media into LAMP)	heat block or qPCR (MD-Bio heater or BioRad_T100 or AB_Veriti or ...)	Twist_MT007544.1 synthetic gRNA	225,000
Aug-31	Mammoth	SARS-CoV-2 DETECTR Reagent Kit	N	Human RNaseP	CRISPR Probe	Automated (Qiagen_955134 on Qiagen_EZ1AdvancedBen chtop)	ABI7500	SeraCare_AccuPlex_0505-0168 IVT encapsulated RNA	60,000
Aug-13	Pro-Lab	Pro-AmpRT SARS-CoV-2 Test	RdRP	only external controls	Probe	Direct (swab into 0.1ml) or kit (swab into 1ml then Pro-Lab_PLM-2000)	Optigene_GenieHT	BEI_NR-52287 inactivated virus	125*
Jul-09	UCSF/Mammoth	SARS-CoV-2 RNA DETECTR Assay	N	Human RNaseP	CRISPR Probe	automated (Qiagen_955134 on Qiagen_EZ1)	ABI7500	SeraCare_AccuPlex_0505-0126 (lot 10480311) IVT RNA	60,000
May-21	Season (1 duplex)	AQ-TOP COVID-19 Rapid Detection Kit	orf1ab	Human RNaseP	PNA Probe	manual (Qiagen_60704)	BioRad_CFX96 or ABI7500	NCCP_43326 gRNA	21,000
May-18/ Aug31	Color	Color Genomics SARS-CoV-2 RT-LAMP Diagnostic Assay	N, E, nsp3	Human RNaseP	Colorimetric	automated (PerkinElmer_CMG-1033 on PerkinElmer_Chemagic360)	plate reader (Biotek_NE02, Hamilton_Star)	ATCC_VR-1986D (gRNA)	2,250
May-06	Sherlock BioSci	Sherlock CRISPR SARS-CoV-2 Kit	orf1ab, N	Human RNaseP	CRISPR Probe	Manual (ThermoFisher_12280050)	heat block or qPCR, Plate reader (BioTek_NE02)	gRNA	20,250
Apr-10	Atila BioSystems	IAMP COVID-19 Detection Kit	orf1ab, N	human GAPDH	Probe	Direct (swab into 350µl, 15min RT then 3µl to LAMP)	BioRad_CFX96 or ABI7500 or Roche_LightCycler480II or Atila_PG9600	SeraCare_AccuPlex_0505-0129 pseudovirus (recombinant alphavirus)	3,500*

Just in 2020!

from Matt McFarlane
(twitter @mattmcfar)

Saliva Direct "most unique EUA" – FDA

- Is a protocol not a product
- Open Access - any CLIA lab can use
- Lots of clone assays being used
- Yale can "designate" CLIA labs

Yale SCHOOL OF MEDICINE

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SalivaDirect: What You Need to Know About the New COVID-19 Test

September 03, 2020 by Michael Greenwood

Twitter Facebook LinkedIn Email Print

About our Test Development

SalivaDirect™ has moved!

For information regarding SalivaDirect™, including lab designation and up-to-date validation results, please visit our new page at <https://publichealth.yale.edu/salivadirect/>.

COVIDTrackerCT will remain home to our surveillance, open-source protocols, and research (including test development).

Saliva

Proteinase K

Combine & mix

Heat inactivate 95°C

Dualplex RT-qPCR

Created with BioRender.com

SalivaDirect: Lab authorization request form

SalivaDirect received Emergency Use Authorization (EUA) from the Food and Drug Administration (FDA) on August 15th, 2020.

The SalivaDirect FDA EUA is for a Laboratory Developed Test, but we have the right to designate other labs its use.

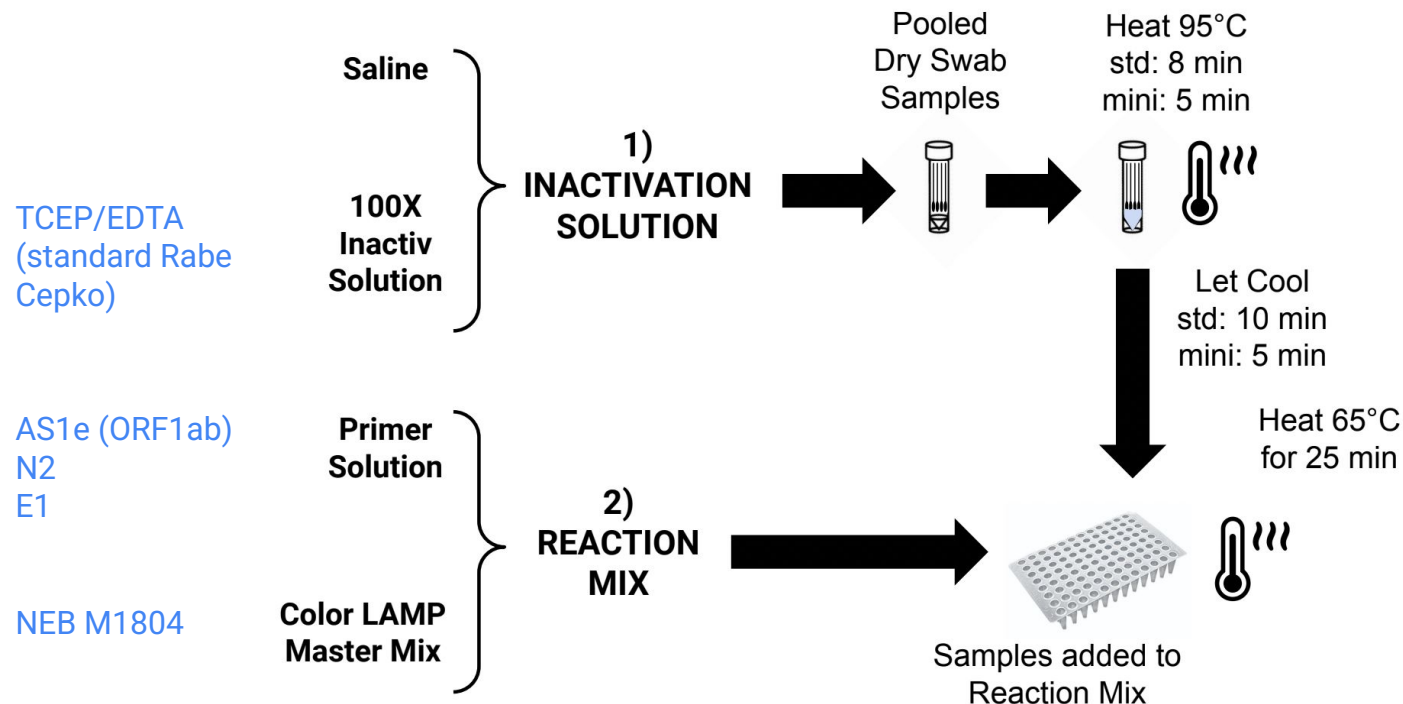
Only high complexity CLIA-certified labs in the United States can become authorized to run SalivaDirect. Please fill out this form if you would like to receive more information on how to become a designated lab.

Do you have access to at least 5x SARS-CoV-2 positive and 5x negative saliva samples for CLIA verification? *

Would you be open to collaboration on bridging studies for your automation systems?

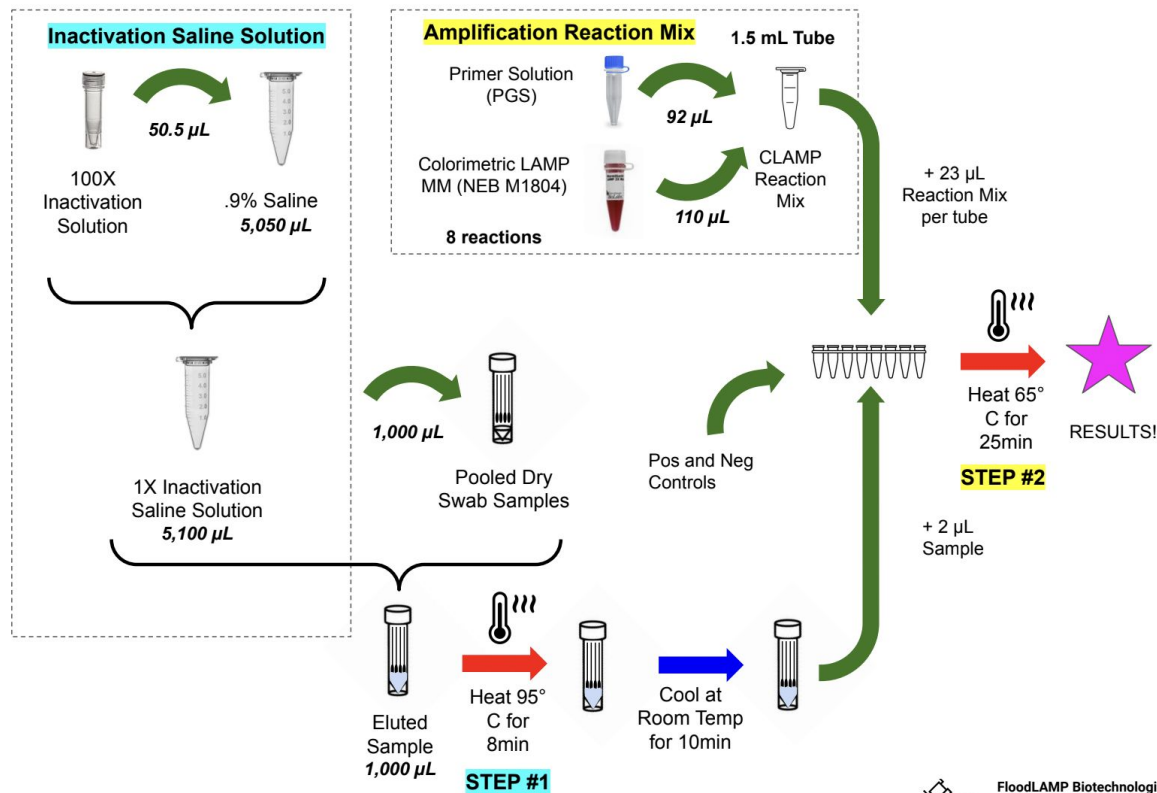
Test Diagrams

FloodLAMP QuickColor™ COVID-19 Test



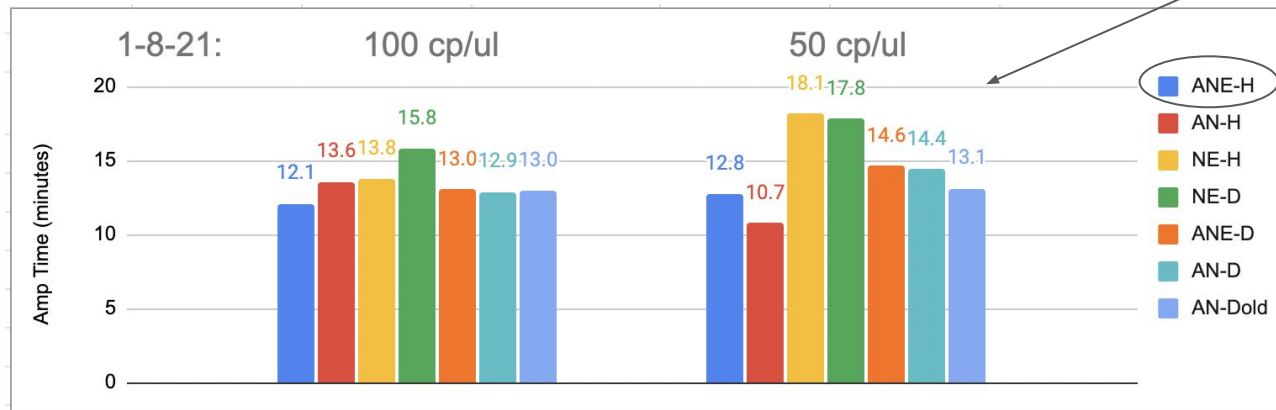
Test Diagrams

FloodLAMP QuickColor™ COVID-19 Test



Primer Comparison

All 3 primer sets chosen by HPLC purification



Run with NEB E1700
Fluorimetric LAMP on
QuantStudio7

*Sample is contrived positive with
Gamma BEI spiked into negative
clinical sample*



A = AS1e Rabe-Cepko primer
set for ORF1ab

N = N2 primer set

E = E1 primer set

"H" = HPLC purified

"D" = Desalted

Successful Commercial Pilots



Commercial deployments in 3 states.



On-site, rapid mass molecular testing performed and operated by EMS staff.



2 cities have opted for mandatory FloodLAMP testing. Prioritizing it over antigen and PCR alternatives.

“

“FloodLAMP’s testing program is a total game-changer. I didn’t know this was possible. I’ve been blown away.”

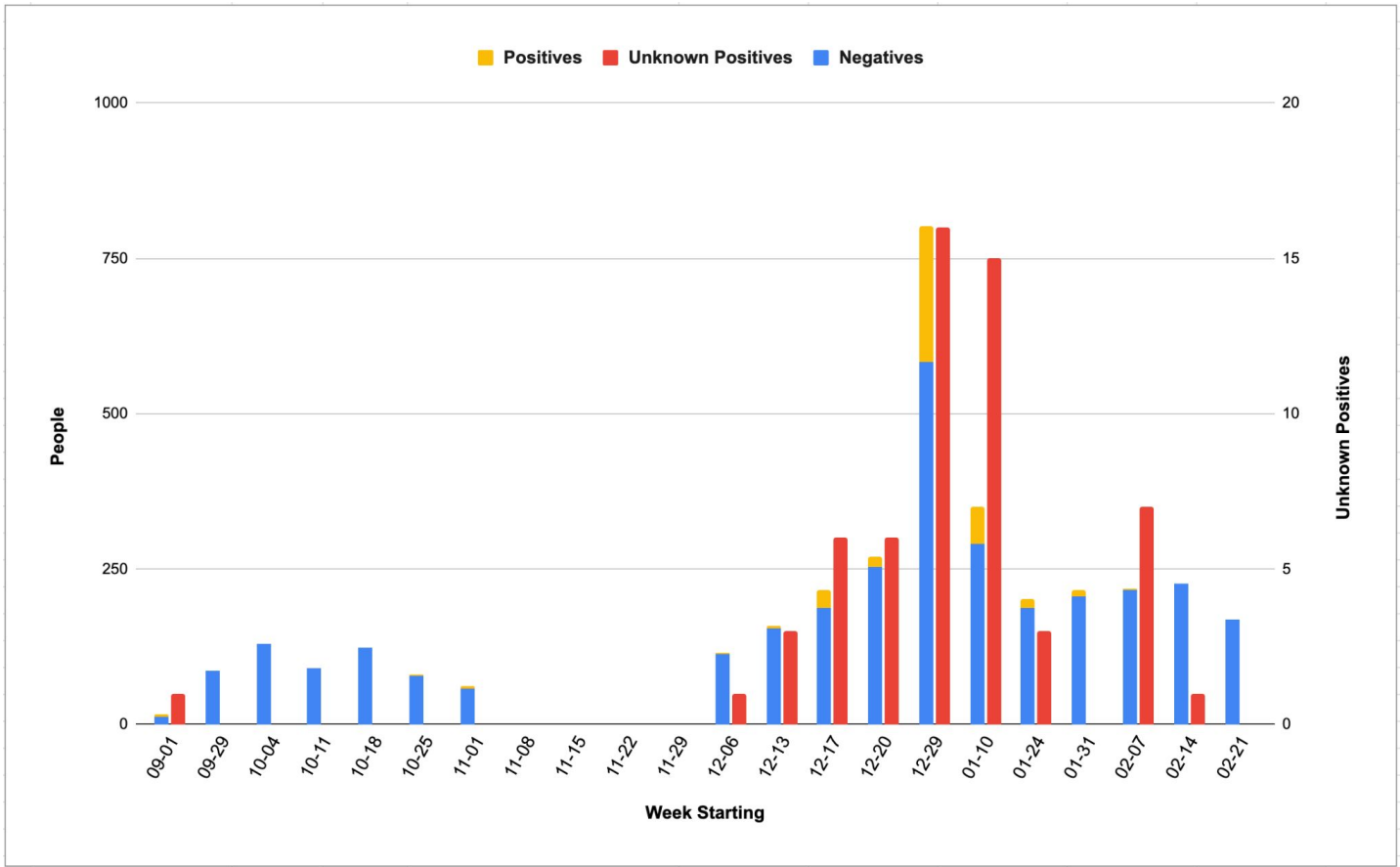
– *President of Production Company for FloodLAMP Pilot program “ROSA”*

"It's been a godsend!"

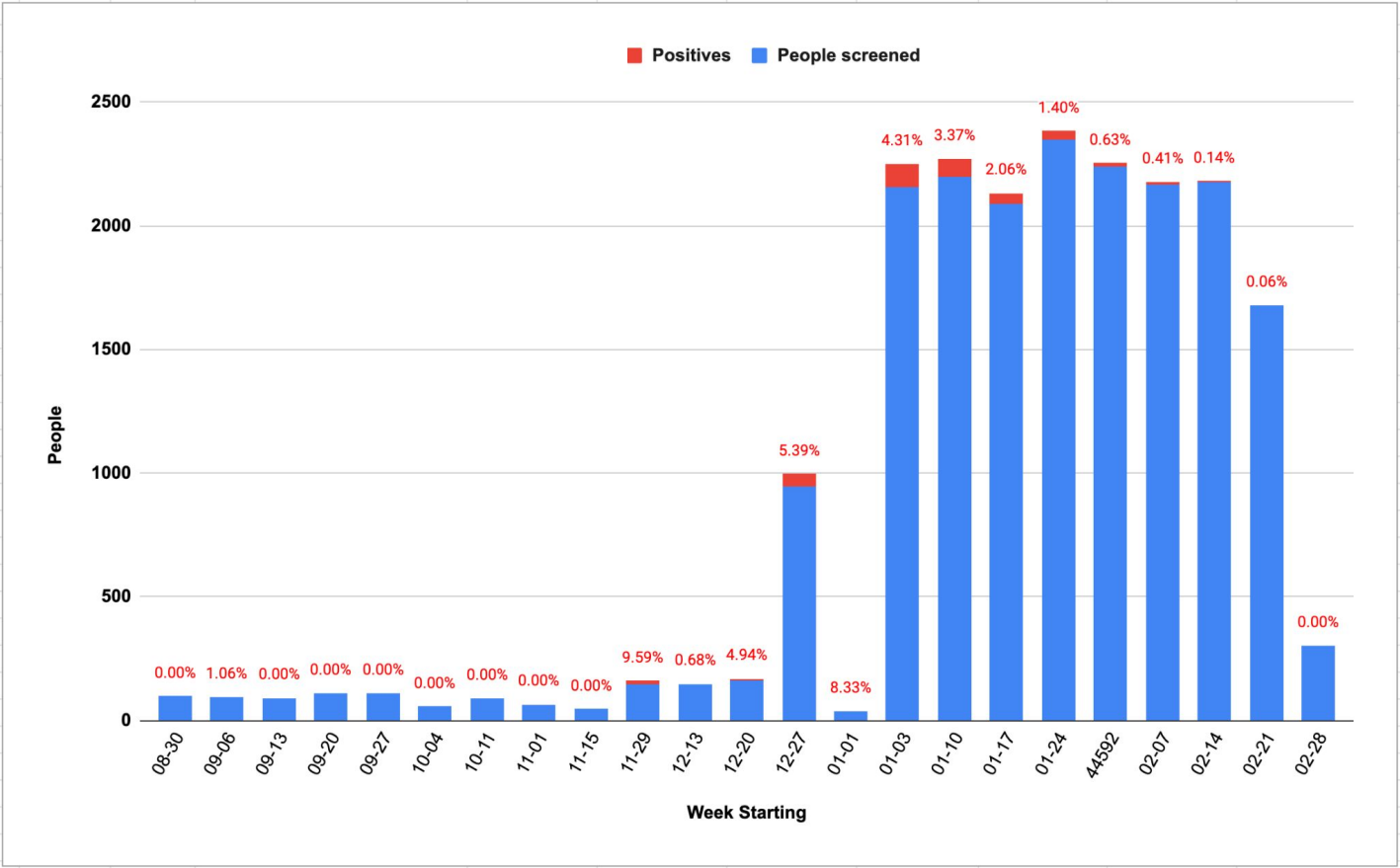
– *EMS Training Director running the FloodLAMP program*

”

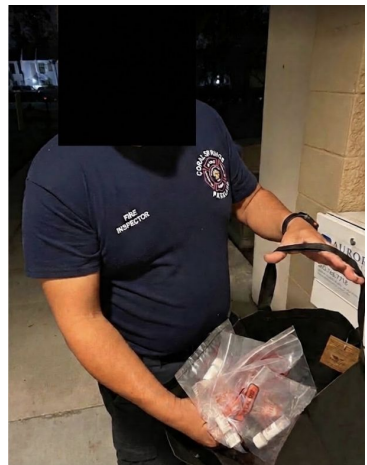
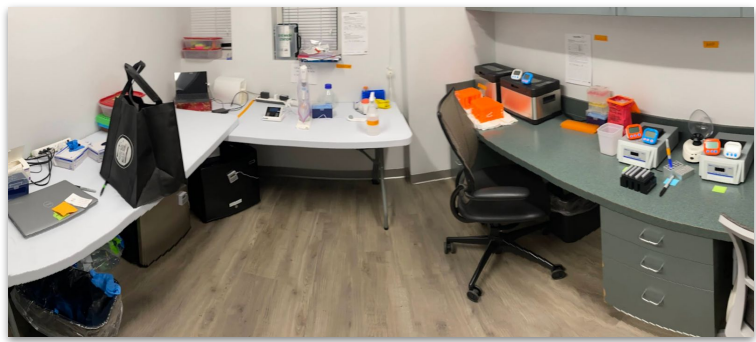
Town of Davie, FL



Coral Springs, FL



Coral Springs, FL

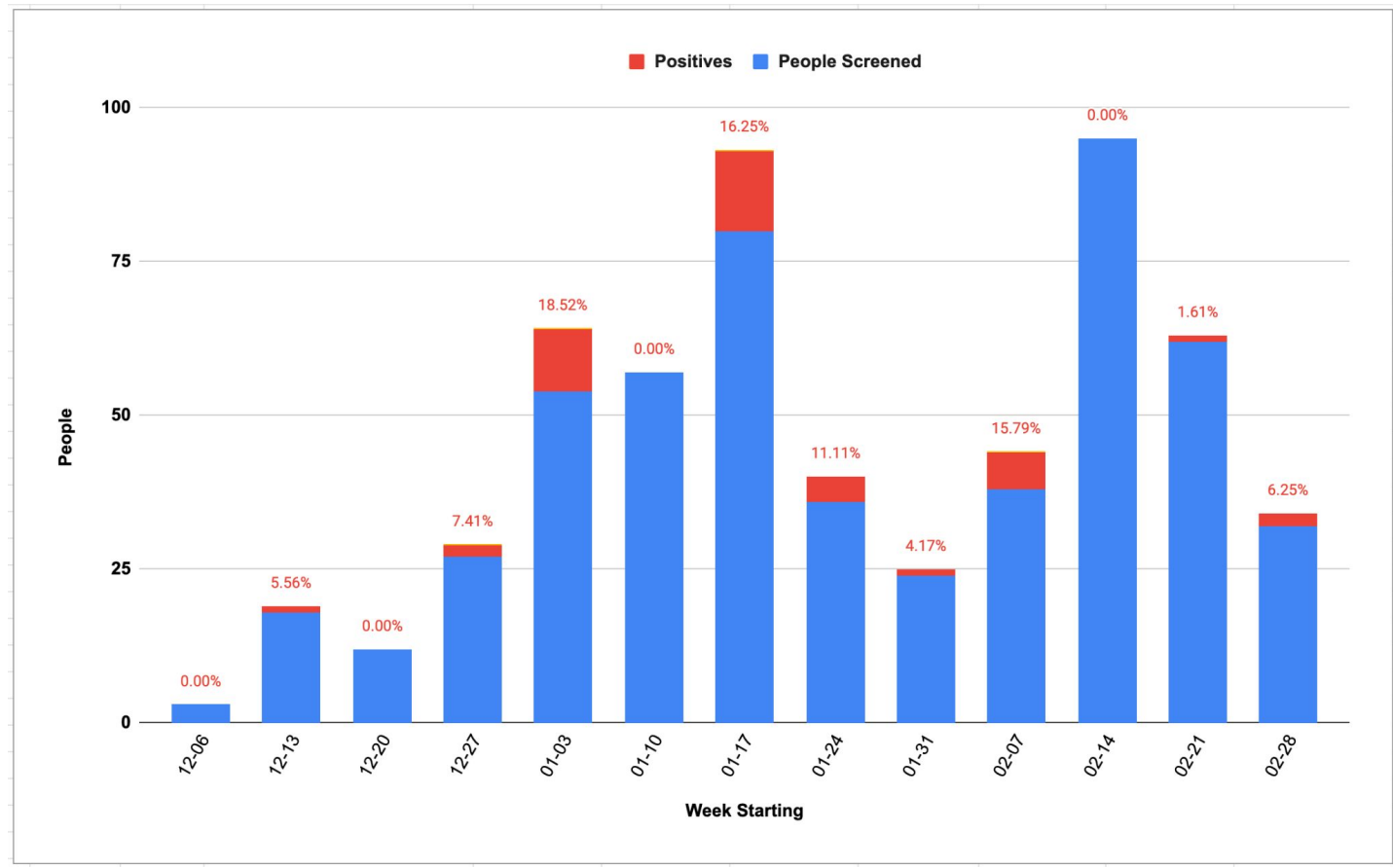


A novice (but dedicated) tester is able to screen 1000 critical city employees.

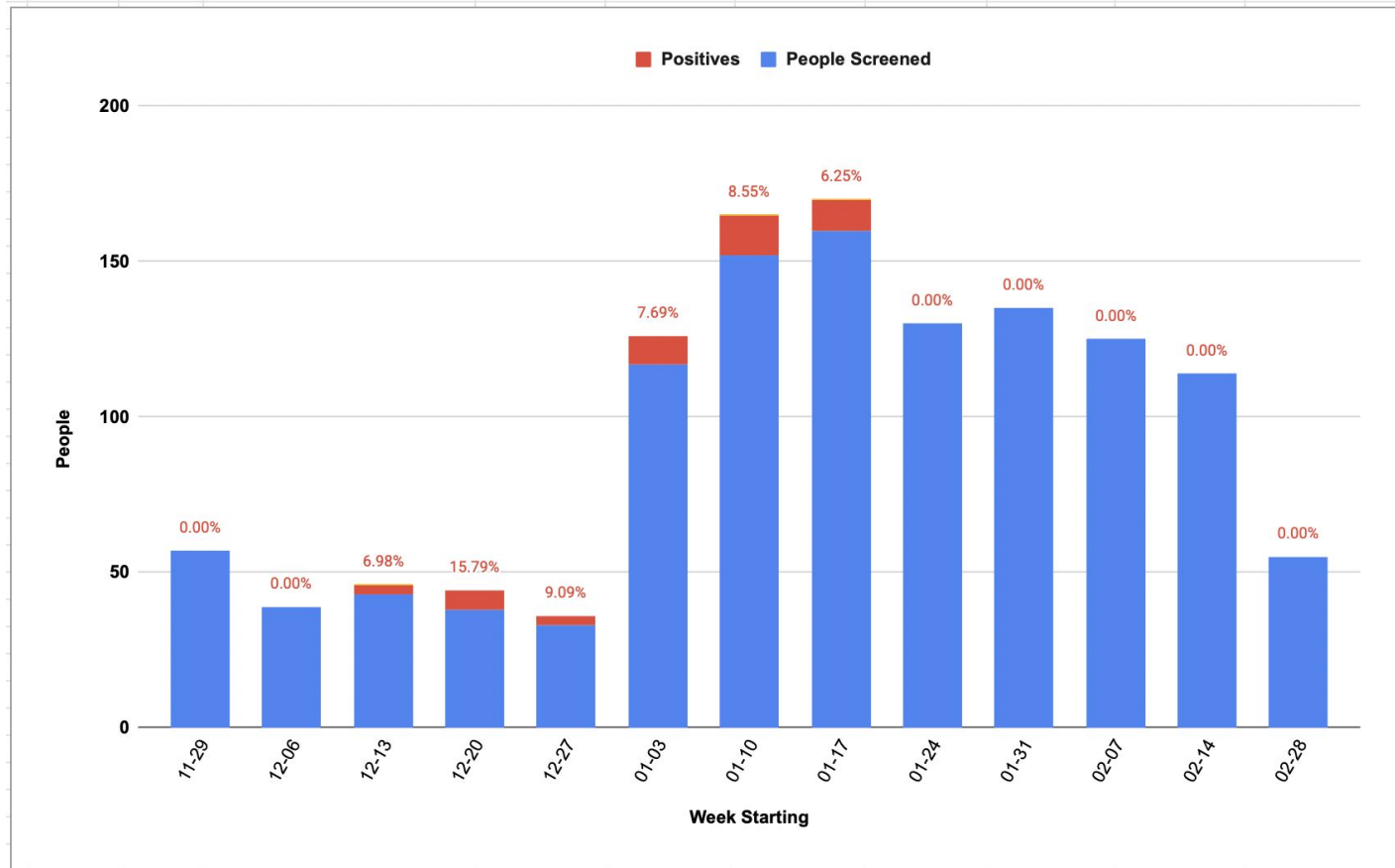
Davie, FL



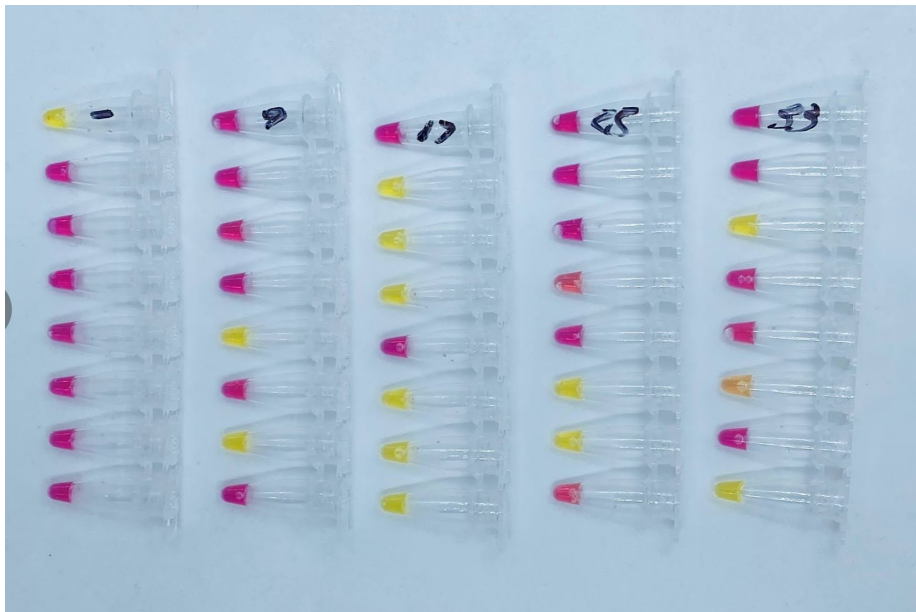
Bend, OR



Preschool, CA



Pre-School Screening



Mon 1-10

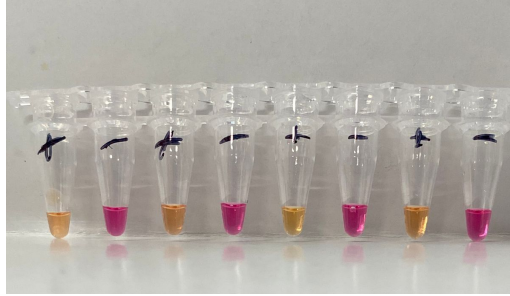


Thurs 1-13

Mix of self collected individuals and family pools

Collaboration with University of Antioquia (Medellin, Colombia)

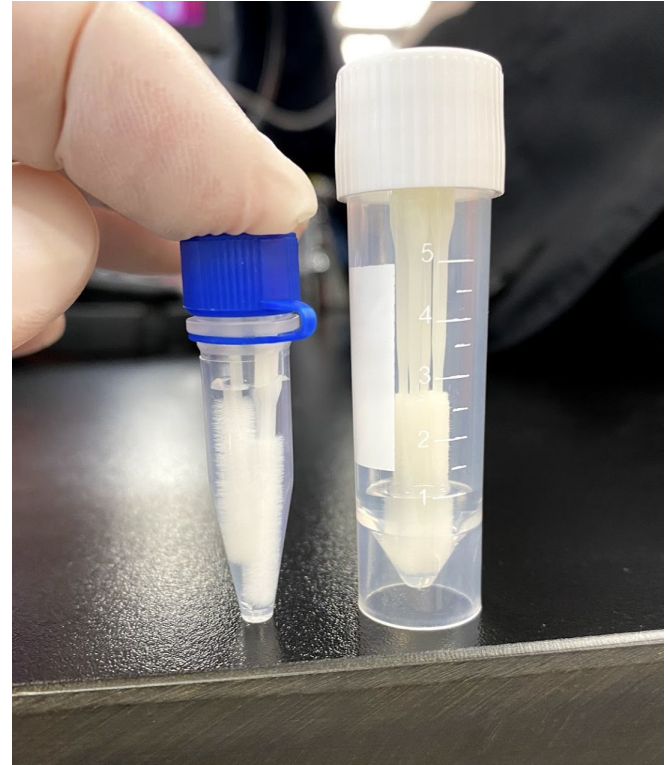
Joint NEB /
FloodLAMP
project



Reagent Kits



Collection Kits



Collection Instructions



**FloodLAMP
Biotechnologies**
A Public Benefit Corporation

FloodLAMP Pooled Swab
Collection Kit

For use with the FloodLAMP
Mobile App



Open your phone camera
and scan to see these
instructions online

FloodLAMP.bio/collectionkits

1  **Begin collection**
Sponsors: Tap the "Collect" button from the app home screen. Follow the instructions in the "Collect your samples" screen, which are listed here below.

2  **Sanitize**
Wash and dry your hands before collecting your sample, or use hand sanitizer.

3  **Open the kit**
Sponsors: Open your kit and distribute swabs to each participant who will be adding samples to the tube.

4  **Open swab packs**
Each participant should open their swab pack. Be careful not to touch the absorbent end with your hands.

5  **Swab the first nostril**
Each participant should insert the swab tip into their first nostril until the absorbent end is no longer visible. Swab the sides of the nostril in a circular motion 4 times. Make sure to twist the swab as you go.

6  **Swab the other nostril**
Each participant should repeat the previous steps with the same swab in the second nostril.

7  **Put swabs in tube**
Each participant should snap off the end of the swab at the break point. Be careful not to touch the absorbent end with your hands.

8  **Cap tube**
Sponsors: Once each participant has added their swab, securely tighten the cap onto the tube. Tap "Next" at the bottom of the "Collect your samples" screen.

9  **Scan tube**
Sponsors: On the "Scan" screen, capture the code on the side of the collection tube. Tap "Next" to continue.

10  **Add participants**
Sponsors: On the "Add Participants" screen, add the names of the pool participants (including the Sponsor if they swabbed) by searching for their names or FloodLAMP registered email addresses. You can add up to 4 swabs in each tube. Participating minors will be registered under their guardian's account.

If you have trouble locating a participant in the system, add their name in the "Collection Notes" field and make sure they sign a consent form.

11  **Put the tube in the biohazard bag**
Securely close the zippered bag, then thoroughly wash or sanitize your hands.

12 **Return the biohazard bag**
By collection point: If your kit does not include a return mailer, bring your biohazard bag to the collection point designated by your program administrator.

By mail: If your kit includes a return mailer, seal the biohazard bag in the return mailer. This packaging is specially designed for carrying samples suspected of carrying infectious substances. Do NOT replace it.
Note: Your return shipping has already been paid.

13  **Register your return**
Sponsors: Tap the "Return" button from the app home screen.

Training Program – Interactive Video

 FloodLAMP ENGLISH (EN) ▾

Quick Color LAMP Training

[Dashboard](#) / [My courses](#) / [QCLT](#)

Welcome

 [Welcome!](#)
To do: [View](#)

 [General Information](#)
To do: [View](#)

Interactive Video Training

 [Instructions and Expectations for Interactive Video Training](#)
Done: [View](#)

 [If you have trouble with EdPuzzle](#)
To do: [View](#)

 FloodLAMP ENGLISH (EN) ▾

5) Reaction Mix Prep

Done: [View](#)





00:00 15:36

◀ ▶ ⏮ ⏭ ⏯ ⏸ 🔊 🗑

◀ 4) Inactivation Reaction

[Jump to...](#) ▾

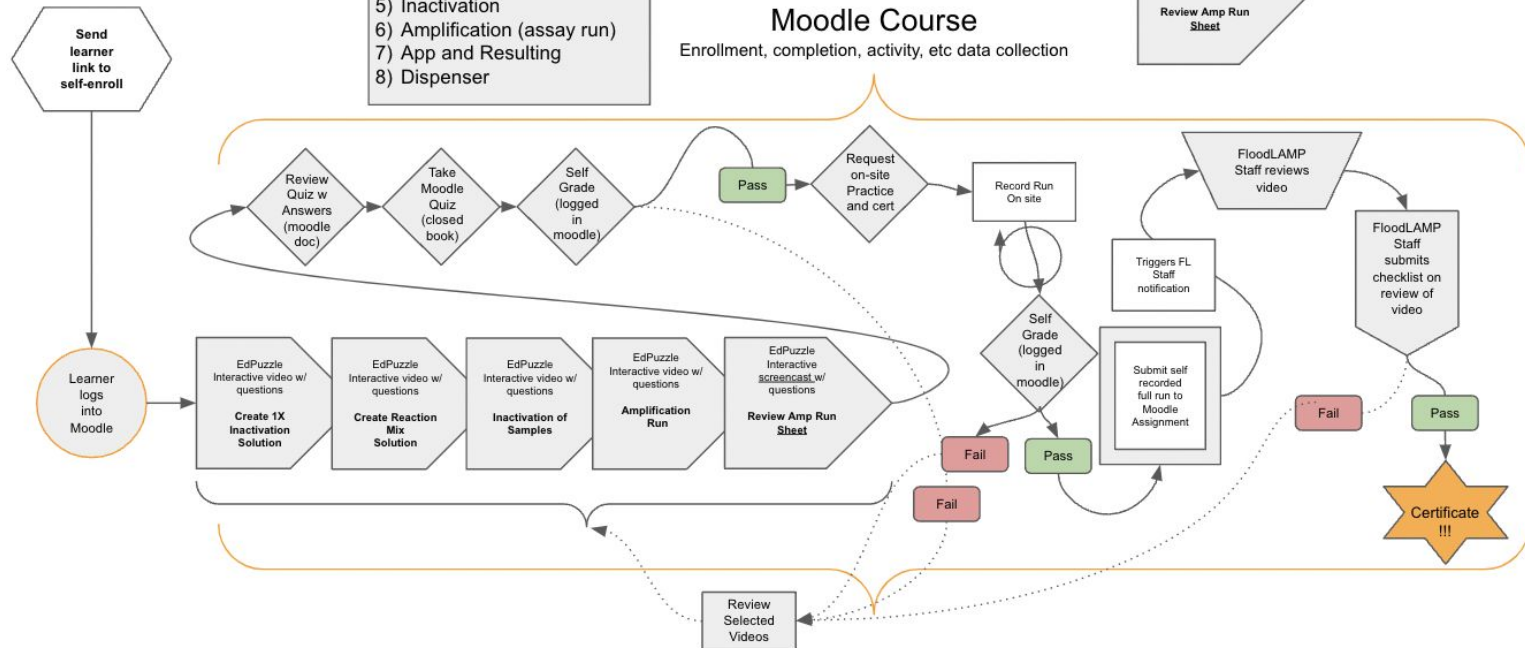
Training Program – Assessment for Certification



Video List for Phase1

- 1) Overview
- 2) Safety and Contamination
- 3) Prep 1X ISS
- 4) Prep Rxn Mix
- 5) Inactivation
- 6) Amplification (assay run)
- 7) App and Resulting
- 8) Dispenser

Moodle Course
Enrollment, completion, activity, etc data collection



Quality Management System

	Document Number:	SOP-102-A_1
	Effective Date:	01/22/2022
	Version:	1
	Reaction Mix Prep - Table Form	

CLAMP MM - NEB M1804 LOT: EXP:	
New In Use CLAMP MM Name: Location: Date and Time:	Retire Link Name: Date and Time:

Reaction Mix Prep

- 1) Thaw PGS and CLAMP MM
- 2) Vortex PGS and CLAMP MM for 3s, centrifuge/shake down
- 3) Add CLAMP MM to PGS tube (or new 1.5mL tube for custom amount)
- 4) Vortex for 10s and centrifuge/shake down
- 5) Add 23µL of reaction mix per well (at bottom, no blowout, no tip touch)
 - * optional use reservoir and multichannel
- 6) Check volumes (look at pink liquid) by lifting plate or strip tubes out of rack
- 7) Label with Reaction Mix ID

Name	Date + Time	Reaction Mix ID (RM_MMDD-1)	CLAMP MM TubeID (_MMDD-X)	PGS LOT# + EXP	Num Rxns	PGS µL	CLAMP MM µL
					48	550	655
					8	92	110

Version 1 SOP-102-A Reaction Mix Prep - Table Form Effective 22/01/2022

Confidential Do Not Distribute without Permission

Page 1 of 2

	Document Number:	SOP-103-A_1
	Effective Date:	01/22/2022
	Version:	1
	Amplification - Run Form	

PREP

- ☐ Heaters on (Heat indicator lit!)
- ☐ 1X Inactivation Saline Soln ready
- ☐ Reaction mix strip8/plates ready
- ☐ Safety Procedures:
 - lab coat
 - gloves
 - face mask
 - face shield or goggles
- ☐ Alcohol wipe sample tubes

+

INACTIVATION

- ☐ Add 1mL of 1X Inactiv Saline Soln to each sample tube
- ☐ Vortex 10sec (tubes) or 30sec (rack)
- ☐ Water Bath (set 99°C)
- OR
 - Heat 8 min then Cool 10 min
- ☐ Dry Heat Block (set 95°C)
 - Heat 5 min then Cool 5 min

AMPLIFICATION REACTION

- ☐ Thaw Reaction Mix (if frozen)
- ☐ Setup tips to align with reactions
- ☐ Add 2µL from each sample tube (pipet up& down 5X, blowout in liquid, tip touch)
- ☐ Heat 65°C for 25 min, set timer
- ☐ Intake sample tubes in App
- ☐ Put sample tubes in fridge
- ☐ Remove Reactions and Let cool for 1 min before photo
- ☐ Take photo in lightbox
- ☐ Crop photo and apply vivid filter
- ☐ Update results in App
- ☐ Log Run with Form link

Version 1

SOP-103-A Amplification - Run Form

Effective 22/01/2022

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Page 1 of 2

	Amplification Log Form Link Location:
Name:	
Date:	
Time:	

Sample Batch ID (from App):
Inactiv Heater:
1X Inactiv Saline Soln ID:
Dispense: Pipette or Manual Disp or Electric Disp

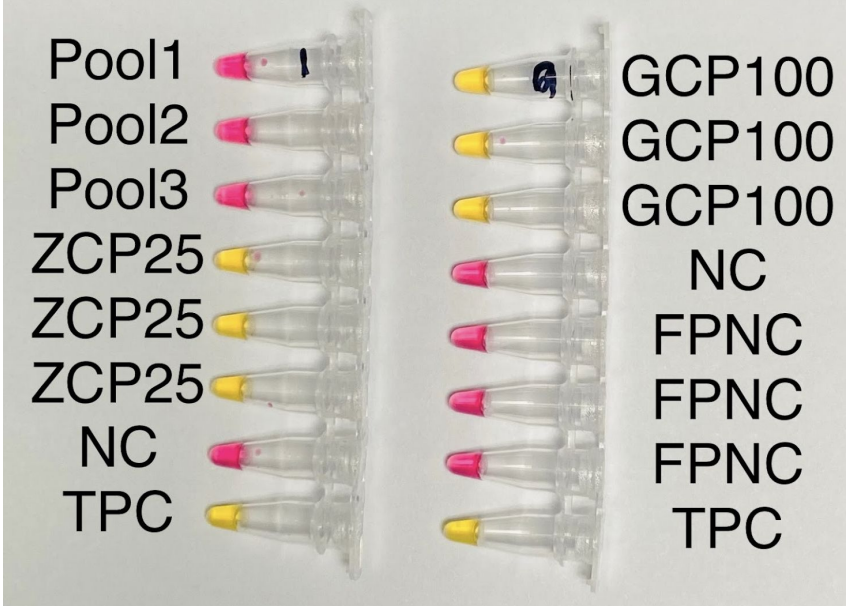
Amp Heater:	Strip8 Tubes or Plates
Reaction Mix: Frozen or Fresh	
Reaction Mix ID:	
Num Reactions (including controls):	
Pos Ctrl ID:	
Neg Ctrl ID (today's 1XISS):	
Initial Inconclusives?:	
Notes:	

→ Implementing ISO 13485 QMS

→ QR Form Driven Logs

New Site Validation Run

FloodLAMP Coral Springs
Validation Run 8-2-21



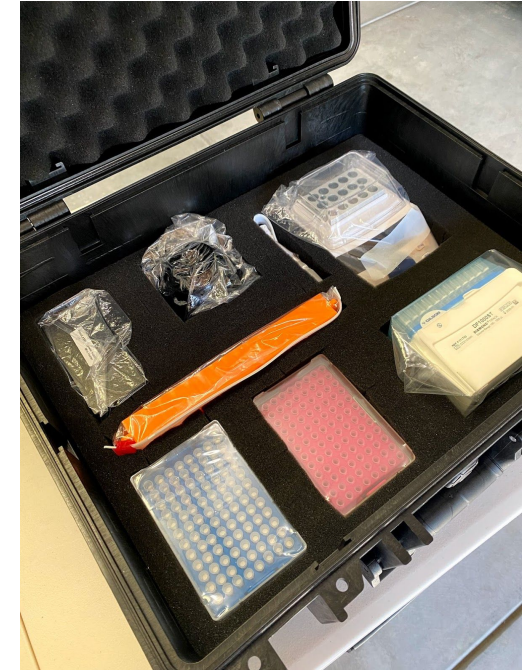
Equipment and Supplies



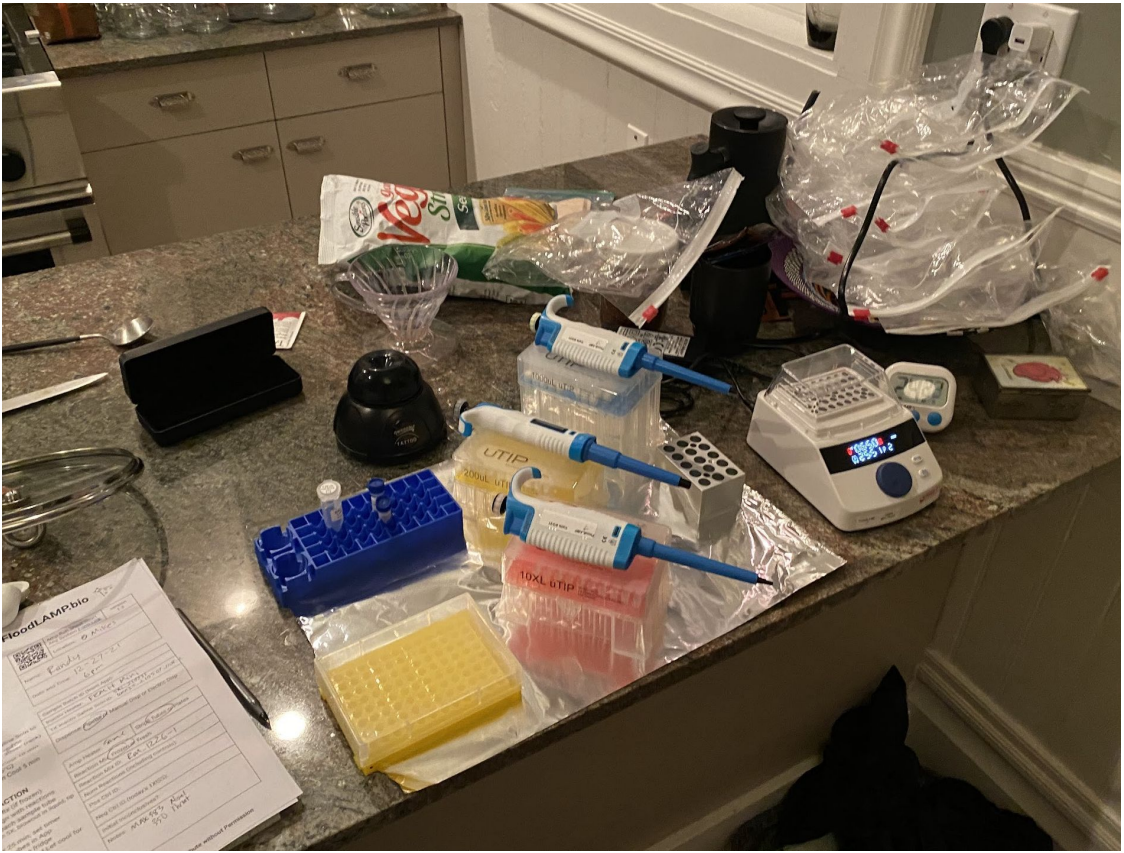
"Lab"



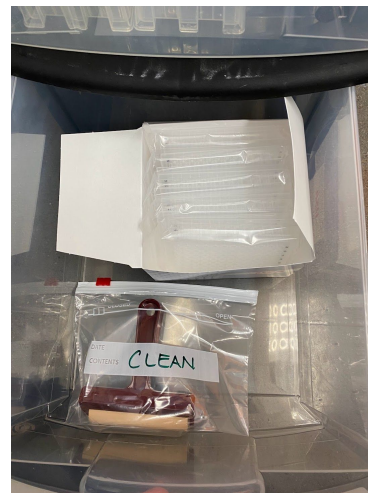
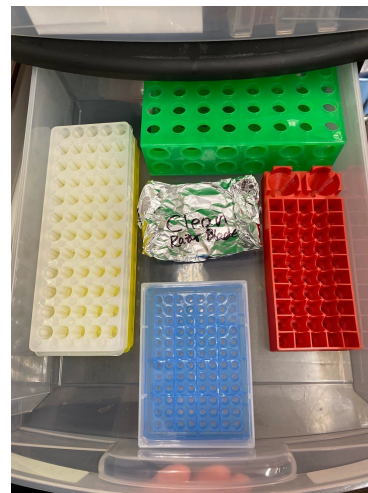
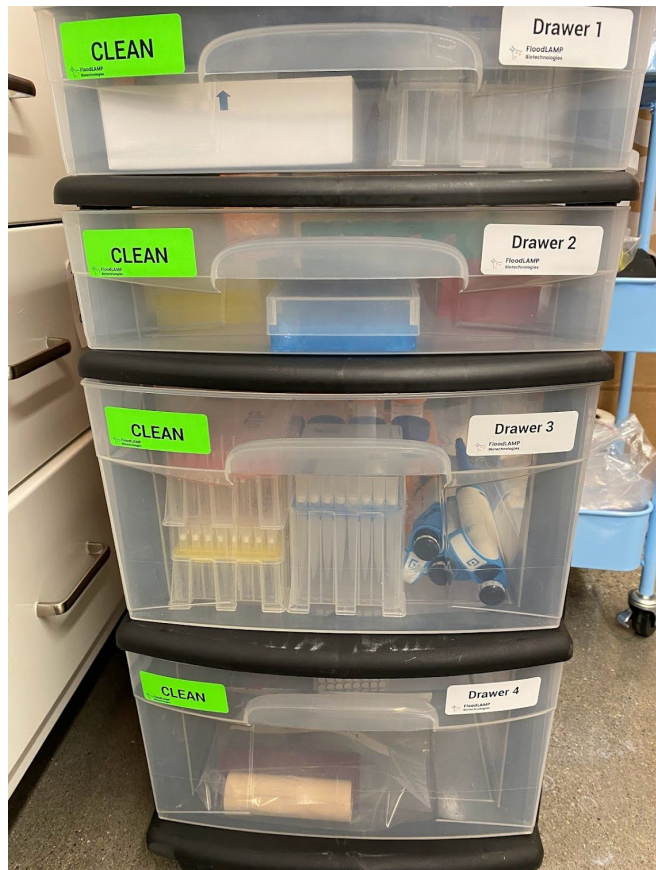
Portable System



Portable System – in action



Prepared Cart

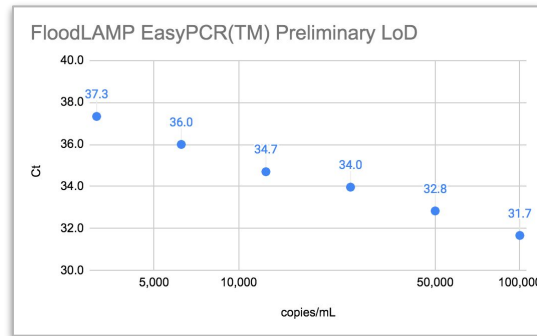


Clinical Evaluation Data

Clinical evaluation performed by the Stanford CLIA Lab, with excellent results and praise on the "really straightforward" protocol.

EasyPCR™ Test

- 3 copies/μl LoD
- 98% sensitivity (PPA 39/40)
- 100% Specificity (40/40)
- No false positives



Gamma inactivated cell lysate from BEI spiked into raw clinical negative sample

QuickColor™ LAMP Test

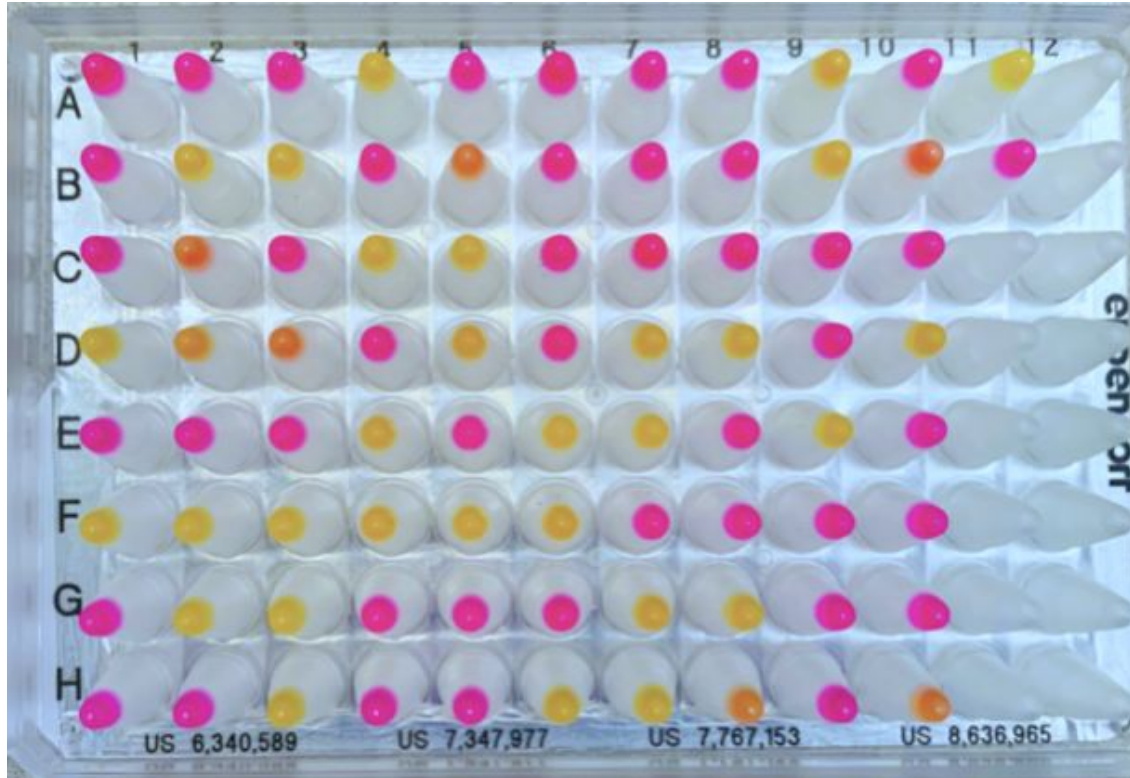
- 12 copies/μl LoD
- 90% Sensitivity (PPA 36/40)
- Missed positives only high Ct (>36 with direct PCR)
- 100% Specificity (40/40)
- No false positives

FloodLAMP SwabDirect PCR Result	Comparator Positive	Comparator Negative	Total
Positive	39	0	39
Negative	1	40	41
Invalid	0	0	0
Total	40	40	80
Positive Agreement	97.5% (39/40) 95% CI: 86.8% to 99.9%		
Negative Agreement	100% (40/40) 95% CI: 91.2% to 100%		

FloodLAMP QuickColor Test Result	Comparator Positive	Comparator Negative	Total
Positive	36	0	36
Negative	4	40	44
Total	40	40	80
Positive Agreement	90.0% (36/40) 95% CI: 76.3% to 97.2%		
Negative Agreement	100% (40/40) 95% CI: 91.2% to 100%		

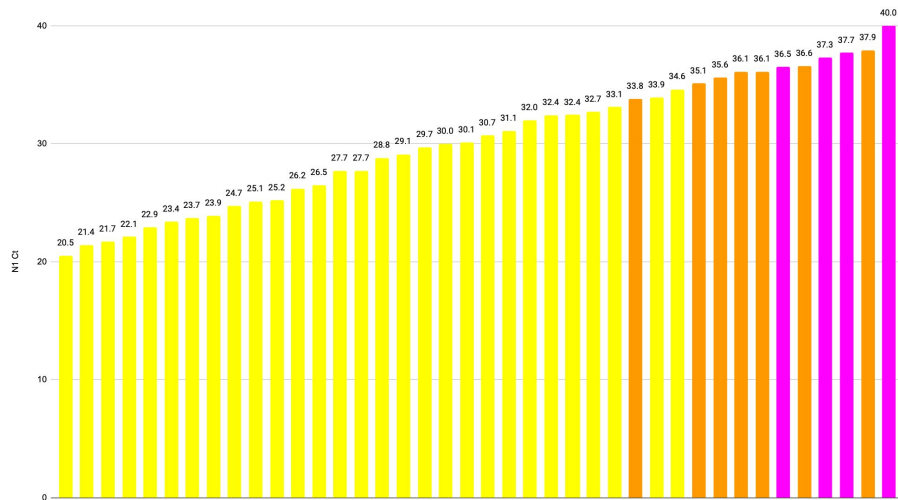
Source of Specimens: Stanford COVID-19 Clinical Testing Program
Specimen Type: Anterior Nares Swab in PBS, previously tested and frozen
Comparator Test: Hologic Panther Fusion SARS-CoV-2 Assay and Hologic Panther Aptima SARS-CoV-2 Assay

Clinical Evaluation by Stanford CLIA Lab



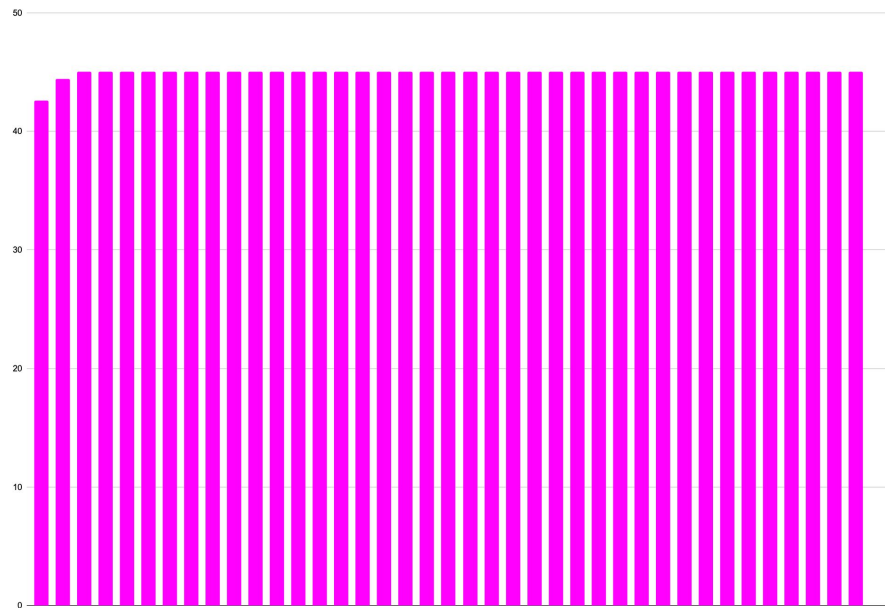
Clinical Evaluation by Stanford CLIA Lab

Direct PCR Ct vs Colorimetric LAMP Result



40 Clinical Positive Samples:

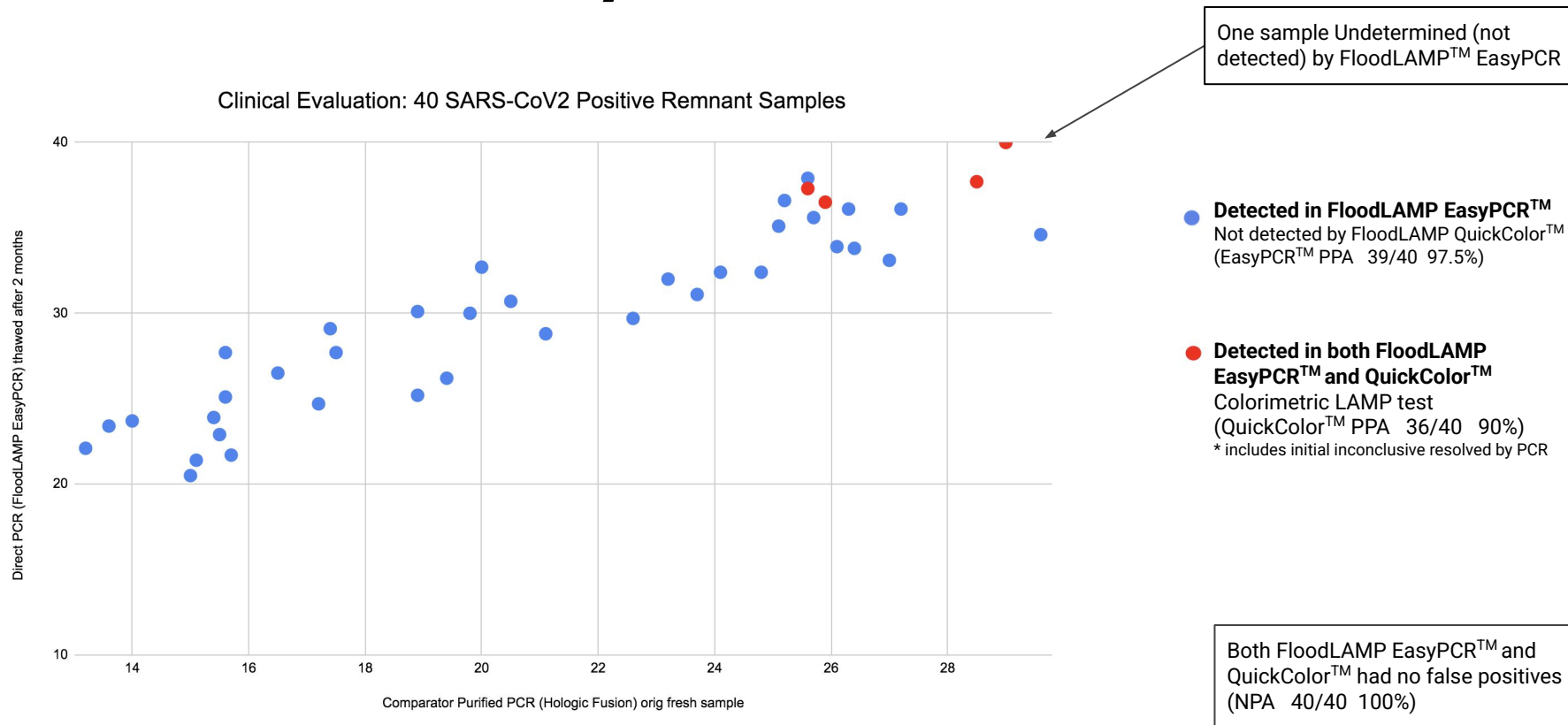
- 29 Positive by FloodLAMP QuickColor™ Test
- 7 Inconclusive (all high Ct, low viral load)
- 4 Negative (all > 36 Ct, very low viral load)



40 Clinical Negative Samples:

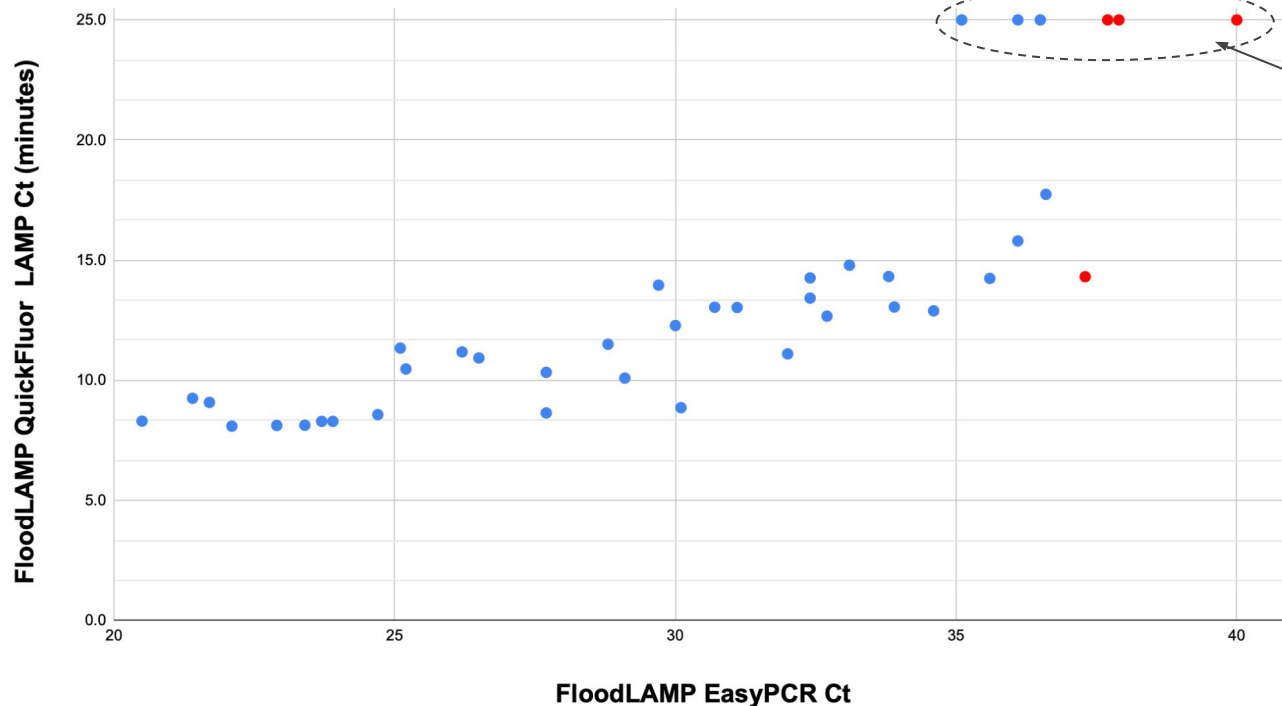
- 40 Negative by FloodLAMP QuickColor™ Test

Clinical Evaluation by Stanford CLIA Lab



Clinical Evaluation by Stanford CLIA Lab

FloodLAMP Stanford Clinical Evaluation - All 3 Tests: Fluorimetric LAMP, Colorimetric LAMP, and PCR



One sample not detected by
FloodLAMP™ EasyPCR
(PPA 39/40 97.5%)

6 Positives Not Detected by
FloodLAMP QuickFluor™ test
(PPA 34/40 85%)

● **Positives not detected by
FloodLAMP QuickColor™ test**
(PPA 36/40 90%)
* includes initial inconclusives resolved by
PCR

All 3 Tests:

No false positives

- FloodLAMP EasyPCR™
- QuickColor™ LAMP
- QuickFluor™ LAMP

(NPA 40/40 100%)

The New York Times

HEALTH

Several common rapid antigen tests work well for Omicron, according to a new study.

The new findings are from an ongoing U.S. study that began in October and was designed to assess the performance of rapid antigen tests in asymptomatic people.

By Emily Anthes

March 1, 2022

medRxiv
THE PREPRINT SERVER FOR HEALTH SCIENCES



BMJ Yale

[View current version of this article](#)

Comparison of Rapid Antigen Tests' Performance between Delta (B.1.61.7;AY.X) and Omicron (B.1.1.529;BA.1) Variants of SARS-CoV-2: Secondary Analysis from a Serial Home Self-Testing Study

 Apurv Soni,  Carly Herbert,  Andreas Filippaios,  John Broach,  Andres Colubri,  Nisha Fahey,  Kelsey Woods,  Janvi Nanavati,  Colton Wright,  Taylor Orwig,  Karen Gilliam,  Vik Kheterpal,  Thejas Suvarna,  Chris Nowak,  Summer Schrader,  Honghuang Lin, Laurel O'Connor, Caitlin Pretz, Didem Ayturk, Elizabeth Orvek, Julie Flahive, Peter Lazar, Qiming Shi, Chad Achenbach, Robert Murphy, Matthew Robinson, Laura Gibson, Pamela Stamegna, Nathaniel Hafer, Katherine Luzuriaga, Bruce Barton, William Heetderks, Yukari C. Manabe, David McManus

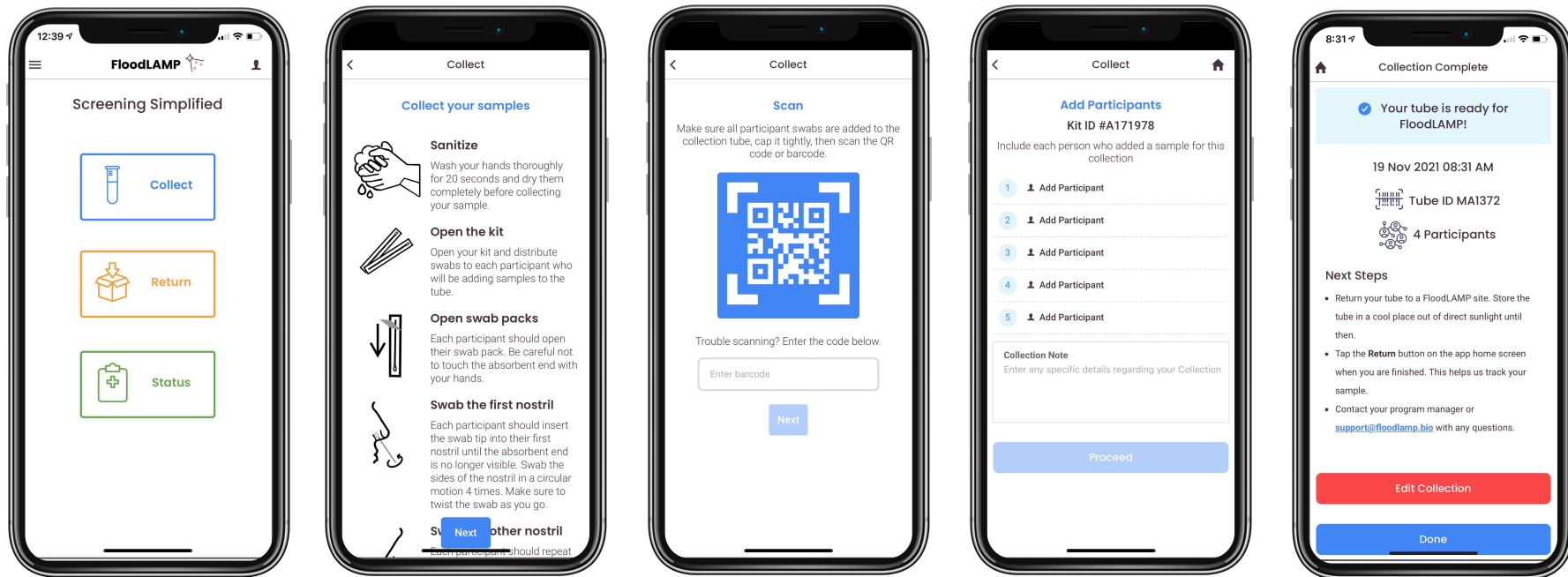
doi: <https://doi.org/10.1101/2022.02.27.22271090>



Results From the 7,349 participants enrolled in the parent study, 5,506 met the eligibility criteria for this analysis. A total of 153 participants were RT-PCR+ (61 Delta, 92 Omicron); among this group, 36 (23.5%) tested Ag-RDT+ on the same day and 36 (23.5%) tested Ag-RDT+ within 48 hours as first RT-PCR+. The differences in sensitivity between variants were not statistically significant (same-day: Delta 16.4% [95% CI: 8.2-28.1] vs Omicron 28.2% [95% CI: 19.4-38.6]; and 48-hours: Delta 45.9% [33.1-59.2] vs.

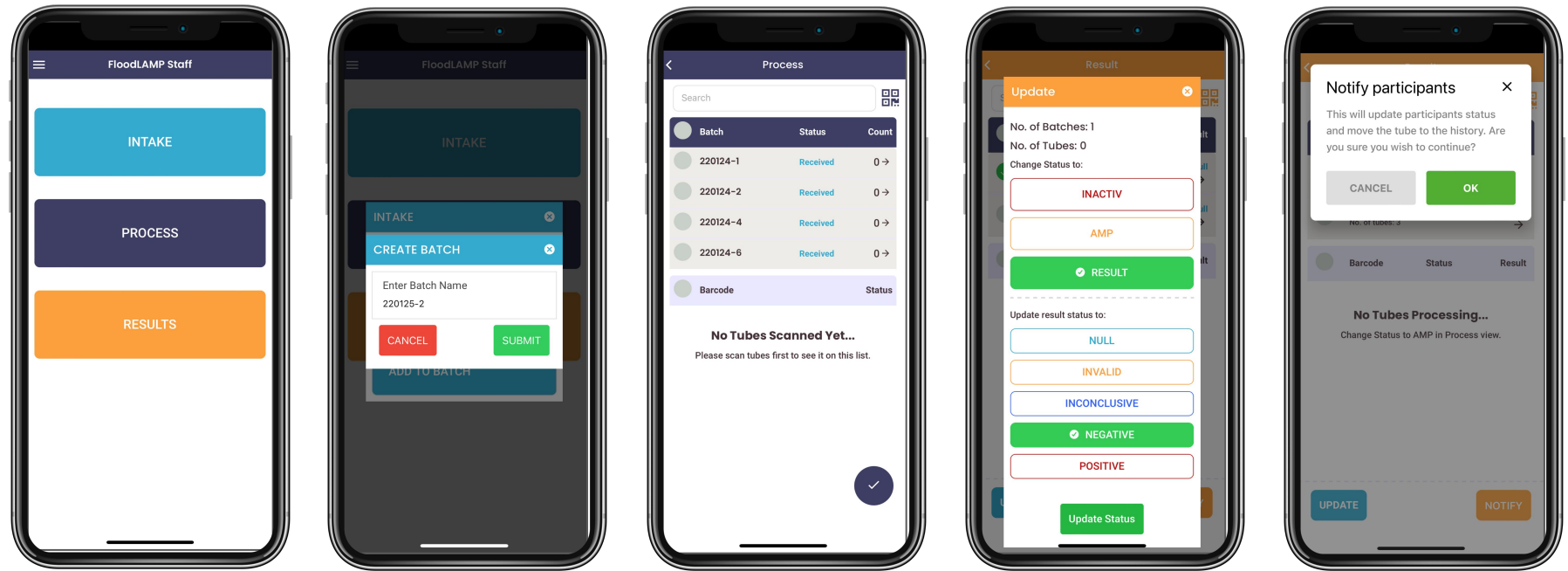
App

For program participants



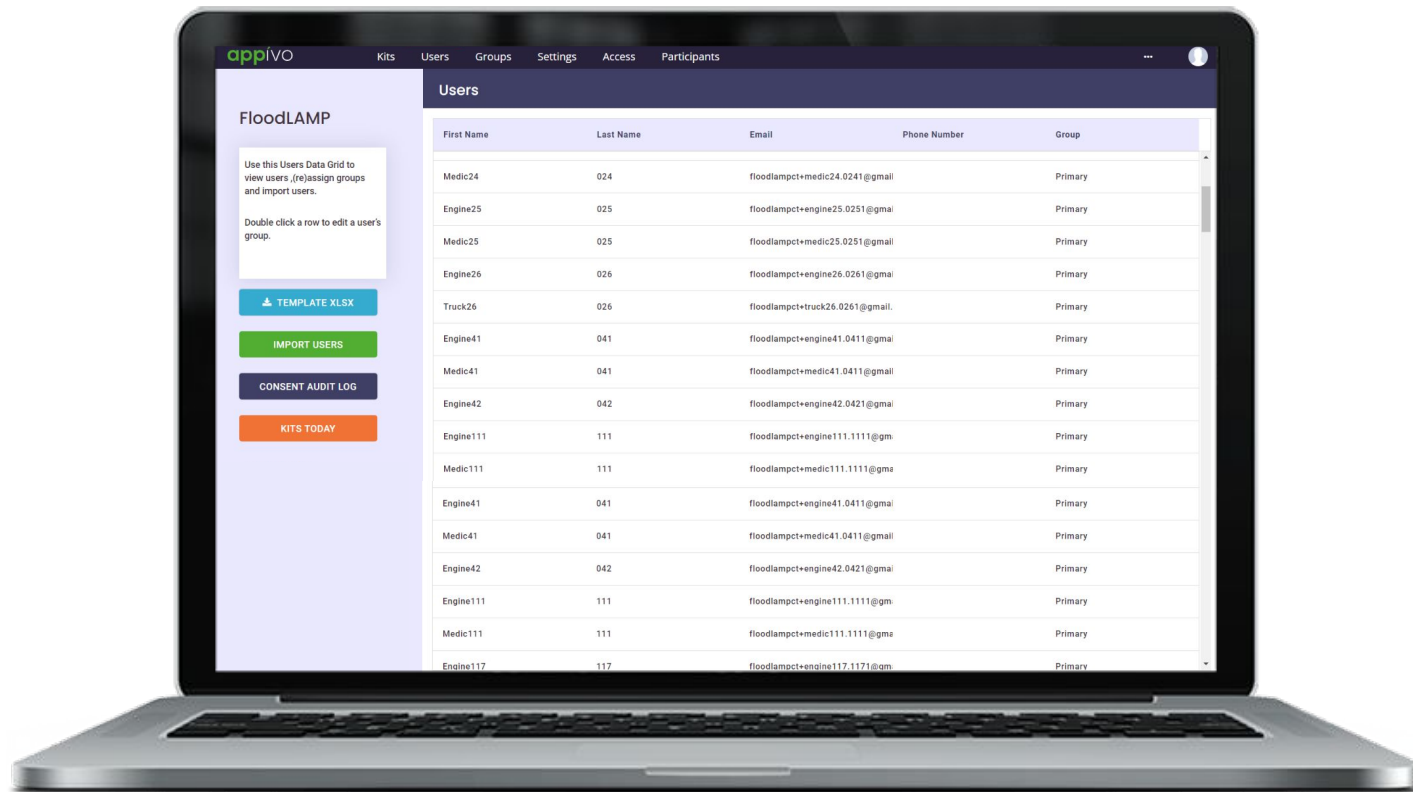
App

For the lab



Admin portal

Tools for program administrators



Team Effort!

